

SMART ENERGY MANAGEMENT

FOR FUTURE NEIGHBORHOODS



Jonas Levi Doll - Battery Specialist

Alex Blandón - Heat Pump Specialist

**Max Concellon Marimon
- Heat Pump Specialist**

Michelangelo Inzerilli - EV Specialist

12 December 2025
MJ2520 - Energy Systems for Smart Cities

Table of Contents

1. Acronyms	4
2. Introduction.....	5
2.1. Stakeholder Overview	5
2.2. Project Goals and Key Performance Indicators	5
2.2.1. Financial Impact	6
2.2.2. Environmental Impact.....	6
2.2.3. Occupant Comfort.....	7
2.3. Description of Strategies Utilized	7
2.3.1. Self-Consumption	7
2.3.2. Price Arbitrage	7
2.3.3. Peak Shaving	8
3. E-Level Analysis: One-purpose strategies	9
3.1. EV Charging Operation Strategies.....	9
3.1.1. Self-consumption	10
3.1.2. Price Arbitrage	11
3.1.3. Peak Shaving	12
3.2. Home Battery Operation Strategies	13
3.2.1. Self consumption.....	14
3.2.2. Price Arbitrage	14
3.2.3. Peak Shaving	15
3.2.4. E-level Battery Strategies Results	17
3.3. Heat Pump Operation Strategies	18
3.3.1. Self-consumption	18
3.3.2. Price arbitrage	19
3.3.3. Peak shaving.....	20
3.3.4. E-level Water Storage Tank Size Sensitivity.....	21
3.3.5. E-level Heat Pump Control Strategies Results	21
4. C-level analysis: Combined Strategies and application on multiple buildings	23
4.1. Batteries Combined Strategies.....	23
4.2. Electric Vehicle Combined Strategies.....	24
4.3. Heat Pump Combined Strategies	25
4.4. Multi-technology control strategy.....	26
4.5. C-level control for 12 buildings.....	27

4.6.	Heterogeneity of Benefits and Impact on Homeowner Satisfaction.....	28
5.	A-level analysis: Integrated community control strategies	29
5.1.	Community level battery control strategies	30
5.2.	Community level EV control strategies	31
5.3.	Community level heat pump control strategies	32
5.4.	Smart Energy Community Results	33
5.5.	Community Governance and Stakeholder Impact.....	34
5.5.1.	Energy Community Governance	34
5.5.2.	Stakeholder Impact	35
6.	Conclusion, final proposal and future challenges.....	36
6.1.	Challenges and disclaimers	37
6.2.	Final proposal	38
7.	References	40
Appendix I.	Financial Inputs and Modelling Horizon	42
Appendix II.	Metrics for tracking objective completion	43
Appendix III.	Sensitivity Analysis of DHW Tank Storage Size.....	45
III.I	Self-consumption (Tank size sensitivity)	45
III.II	Price arbitrage (Tank size sensitivity)	46
III.III	Peak shaving (Tank size sensitivity)	46
Appendix IV.	Heat Pump Control Combined Strategies	48
Appendix V.	Optimal Controller Impact on Individual Neighbourhood Households.....	49
Appendix VI.	Comparison of battery controller results (E&C_Level)	51
Appendix VII.	Different control logic for the HP in each of the 12 households	52

1. Acronyms

BESS – Battery Energy Storage System

CAPEX – Capital Expenditure

DHW – Domestic Hot Water

DSO – Distribution System Operator

EC – Energy Community

EV – Electric Vehicle

HP – Heat Pump

HPT – High Price Threshold

LPT – Low Price Threshold

NPV – Net Present Value

PA – Price Arbitrage

PR – Price Ratio

PS – Peak Shaving

PV – Photovoltaic

REC – Renewable Energy Community

SC – Self Consumption

SH – Space Heating

SOC – State of Charge

TLCC – Total Life Cycle Costs

2. Introduction

In recent years, the clean energy transition has spurred massive growth in decentralized energy generation through residential photovoltaic (PV) energy systems. Accompanied by the rapid electrification of energy demand, such as the adoption of heat pumps (HP) [1] and electric vehicles (EV) [2], residential energy systems in Sweden are greatly transformed. Whilst these shifts create challenges for all stakeholders involved in the energy system, from rising electricity bills to higher grid strain, there are also an array of opportunities to be exploited.

The following report presents a detailed study of energy solutions for a new multi-property development in Sweden, consisting of twelve single family homes that collectively represent a neighbourhood. In line with the latest trends, homes are to be equipped with heat pumps, solar photovoltaics, and will host EVs. Despite being designed with high energy efficiency standards, several improvements can be applied to the development properties, through a smart control of the HP and EVs, and through the installation of a Battery Energy Storage System (BESS). A detailed investigation with *Polysun* modelling software is presented, and with the financial inputs found in Financial Inputs and Modelling Horizon, smart controls for these technologies are assessed on their potential to improve the attractiveness of the property development. Furthermore, the potential to create an energy community is explored, by considering the development of a centralized heating, and a microgrid for the sharing of electricity between neighbours.

2.1. Stakeholder Overview

The development of a new neighbourhood includes several key stakeholders that will be influenced by the strategies proposed and will in turn influence the strategies themselves. The focus of this work is to develop a solution that will improve the conditions for household owners in the new development, which in turn provides a better selling offer to be advertised by the property developer. Consequently, the main objectives and KPIs of the analysis are tracked from the perspective of future residents, who are also considered to be the owners of the houses and any assets located in it, such as EVs. Providers and installers of technical solutions are considered neutral agents.

Additionally, the perspective of the Distribution System Operator (DSO) is considered for the development of an energy community, given its impact on their operations as the provider of grid connections for the neighbourhood. In the latter stages of the report, considerations are made regarding the DSO's interests or otherwise for such an energy community, as opposed to a baseline case of decentralized households connected to the grid in a traditional manner. Finally, regards are placed on legislators, at a municipal and national level, as agents of change that provide regulatory and financial support to steer residential energy management in the direction that will maximize social welfare.

2.2. Project Goals and Key Performance Indicators

The objectives of the smart building strategies implemented in this report are threefold; To minimize the lifetime costs of all electrical systems in the house, minimize operational GHG emissions, whilst maintaining comfortable living conditions for the building occupants. By lack

of a unified method to quantify each of the three goals against each other, the best control strategy is decided as a trade-off between the three main objectives, through subjective interpretation of the results. An overview of the *Polysun* results that can be used to track these objectives can be found in Appendix II in page 43. The specific sub-objectives that are tracked to evaluate different strategies can be found below, together with the custom metrics developed to properly assess each objective's completion.

2.2.1. Financial Impact

To minimize the costs of operating electrical appliances in the building, the Net Present Value (NPV) of any control strategy can be considered, which signals the total costs of operating the system over a given timespan. It must be noted that a higher NPV is analogous to a reduction in Total Lifecycle Costs (TLCC) of the same magnitude. Two main approaches to increasing the system's NPV can be considered.

- Reduce the yearly sum of energy purchases.
 - o Reducing the kWh of energy consumed leads to lower yearly costs, for which the yearly *Electricity Consumption [kWh]* provides an indication on the energy that is being paid for in the annual *Energy Supply Costs [SEK]*.
- Minimize the price of purchased electricity.
 - o Cheaper electricity in the energy supply bill can be tracked through the *Average Cost of Electricity [SEK/kWh]*, calculated as the annual *Energy Supply Costs [SEK]*, divided by the annual *Electricity Consumption [kWh]*, and excludes savings from lower overall energy usage.
 - o Given that an electricity surcharge links to the magnitude of the year's highest power peak, minimising the *Peak Power Import [W]* from the grid has a direct relationship with lower supply costs.

Whilst the ultimate financial goal is to reduce the TLCC, this is achieved through lower supply costs for a similar amount of CAPEX. It is possible to reach this goal through two approaches, and generally reducing the *Average Cost of Electricity [SEK/kWh]* will yield greater savings for a similar level of comfort, for which it is important to consider the trade-offs between this metric and the annual *Energy Supply Costs [SEK]* when modifying the operation of heat pumps.

2.2.2. Environmental Impact

To monitor the environmental impact of the system, *CO2 Emission [kg]* of the system can be tracked, which account for the embedded emissions in grid electricity purchases. Whilst sustainability is a very important metric to study energy usage, the chosen approach is limited in its relevance to Sweden, given that grid embedded emissions are generally lower than the European average considered for this study. Furthermore, alternative approaches exist to count emissions, such as considering credits for clean PV energy exports, in which case the calculated emissions of a building with less than 50 % SC would be negative. Consequently, in this specific study the environmental impact is considered as having the lowest weight in the final qualitative analysis of objective trade-offs.

2.2.3. Occupant Comfort

Excessive manipulation of HP controls and EV charging schedules can lead to the energy needs of house occupants not being met, in which case homeowners would rebuke the proposed strategies. Metrics considered to track occupant comfort are the following, and strategies that don't meet the minimum lower limits are discarded.

- *Energy Deficit [kWh]*
 - o Tracks the difference between energy demand and energy consumption, indicating where the system demand is not met.
- *Minimum building T° [$^\circ\text{C}$]*
 - o Lower limit of 20 $^\circ\text{C}$.
- *DHW tank top layer minimum T° [$^\circ\text{C}$]*
 - o Lower limit of 46.9 $^\circ\text{C}$.

With the energy deficit ($Edef$), building min. T° ($BuildT$), and DHW min. T° ($DHWT$) a global comfort level indicator is developed to compare the baseline (subscript b) with a given case (subscript c), which is crafted according to **Error! Reference source not found..**

$$Comfort\ Change = \frac{Edef_b - Edef_c}{Edef_b} + \frac{BuildT_c - BuildT_b}{BuildT_b} + \frac{DHWT_c - DHWT_b}{DHWT_b} \quad (1)$$

2.3. Description of Strategies Utilized

Three main approaches are taken to achieve the stated goals for smart building control namely maximizing self-consumption, price arbitrage, and peak shaving.

2.3.1. Self-Consumption

The first strategy is to increase PV self-consumption (SC), with the goal to maximize the amount of PV production that is consumed in the building where it is produced and not exported to the electricity grid. By shifting the demand of electrical consumption devices, the excess of PV production can be reduced, and storing it in a battery system for later use further reduces the exports to the grid. Through increases in PV self-consumption, less grid electricity must be purchased for the same energy consumption, which reduces operational expenses given a PV export feed-in price that is lower than grid electricity purchases, and minimizes the indirect emissions associated with the grid electricity mix.

2.3.2. Price Arbitrage

Price arbitrage (PA) aims to take advantage of fluctuations in spot market electricity prices to reduce the total electricity supply costs. This can be done by either shifting consumption to times of low electricity prices and away from high price periods, or by using storage to increase purchases in low price periods to be discharged when needed.

To find out the optimal price thresholds, we investigated the distribution of relative prices throughout the year (Figure 1), calculated for each time t as shown in (2).

$$PRt = \frac{Pt - \min(P_{0-23})}{\max(P_{0-23}) - \min(P_{0-23})} \quad (2)$$

The analysis reveals the lowest and highest 2 % of relative prices occur much more often than prices in between, suggesting a strong variability in prices.

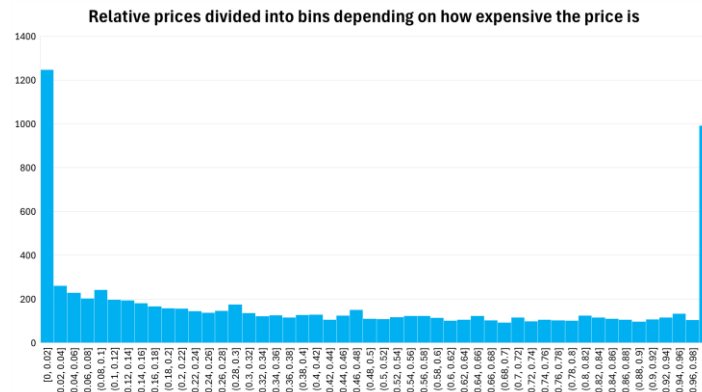


Figure 1: Relative price distribution depending on price values in 2 % ranged

Considering the distribution of absolute prices throughout the year reveals that, even though the prices are very variable, most of the time they don't have big differences in 24 hours, as can be seen on Figure 2. This shows that even though the relative prices are very variable, the absolute differences are mostly kept between 0.1 and 0.8 SEK/kWh, which is relevant to design PA strategies that acknowledge this low spread in prices.

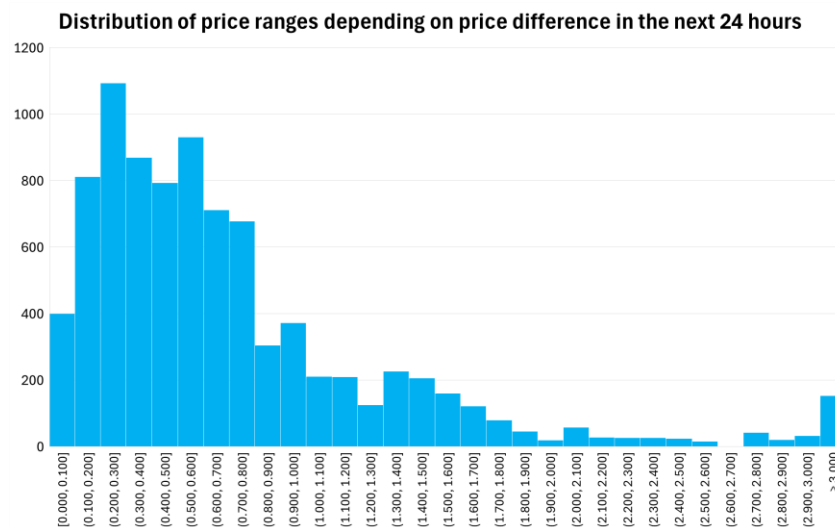


Figure 2: Distribution of maximum price differences during the next 24 hours

2.3.3. Peak Shaving

The last strategy explored is Peak Shaving (PS), which reduces the yearly maximum average hourly power consumption, and therefore minimizes the associated costs that are charged by the electric utility. This strategy can be implemented by shifting loads or using energy storage as a buffer, to reduce power demand at critical times when grid imports are high. In the base case for house 1012, the highest yearly peaks of average hourly grid power consumption can be observed in Figure 3, where the load duration curve is shown for the net consumption of household assets, that is subtracting own PV production, which is not part of the grid imports. The objective of peak shaving is to reduce these peaks, by modifying the consumption of EV, Appliances or HP, at the time of peak power demand.

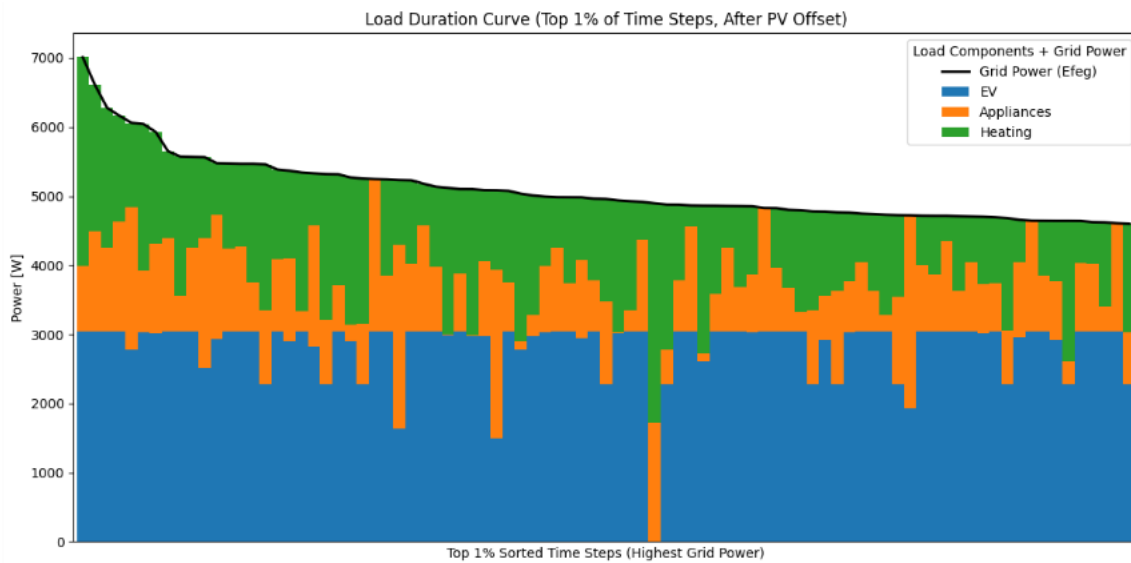


Figure 3: Top 1% load duration curve of the base case for house 1012

3. E-Level Analysis: One-purpose strategies

3.1. EV Charging Operation Strategies

The baseline model used for only EV strategies is the house 1001. The electric vehicle (EV) taken into consideration is a Tesla Model S 75D, with a battery nominal capacity of 70 kWh. The charging station is an AC type 2 Level 2.

The goal of the self-consumption, price arbitrage and peak shaving strategies is respectively to increase the usage of electricity coming from the PVs, to reduce the energy supply costs and to rise the NPV value compared to the baseline, and finally to reduce the peaks of the imports from the grid. These indicators will be analysed individually on each strategy, to later combine the three strategies to achieve the best results for the three main indicators.

The schedule of the EV can be seen in Figure 4.

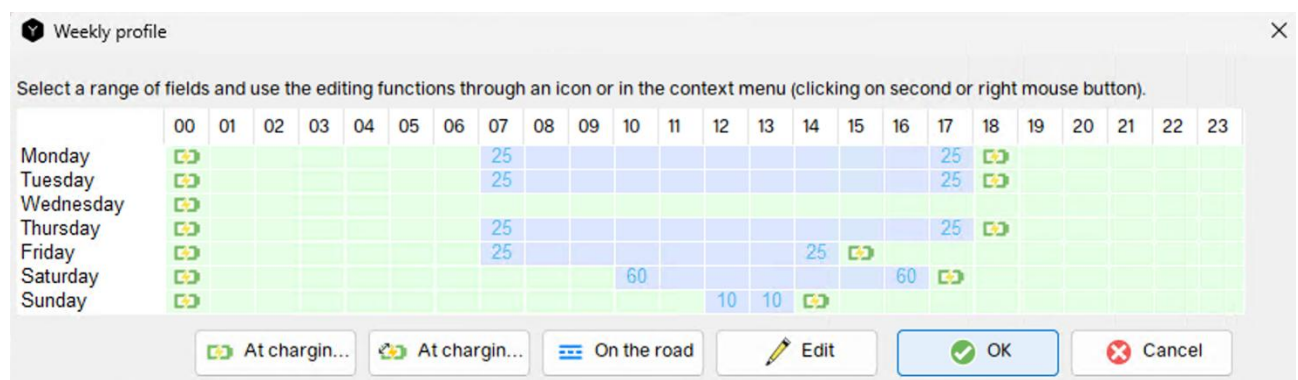


Figure 4: EV Schedule for House 1001

As shown, the vehicle is at the charging station for most of the time over the day, and particularly overnight and for consecutive hours. This information shows how fast charging is not needed, avoiding unnecessary peaks.

3.1.1. Self-consumption

Looking again at the schedule, the EV stays a minimum of 10 hours consecutively at the charging station. This happens on Mondays, Tuesdays, Thursdays and Fridays. This analysis enhances how it not needed to charge at 11 kW (predefined full power), while enriches a more stable distribution of the energy over time. Because of that, a full power option will be chosen by the controller only when enough PV's energy is available.

Full power charging is permitted only under State Of Charge (SOC) of 50 %, and a general maximum percentage of charge is set to 80 %, to preserve the battery's health.

Different strategies have been tried, varying the number of the levels of power outputs, together with their values and including or not boundary of having a net self-production from the PVs higher than zero. Three different power levels strategies comprehend power outputs of: 11 and 3 kW for the first strategy, 9, 3 and 1 kW for second, 9, 3 and 2 kW for the last one. Using only 1 kW when there's not enough production from the PVs results in not charging enough the vehicle, which can't ride over the predefined kilometres. Lowering the amount of power taken from the grid has the aim of leaving "more space" to the power which can be used from the PVs later, when their production will increase, then rising the self-consumption.

The best performing strategy charges at a full power (9 kW) until SOC of 50 %, followed by a partial power of 3 kW when the SOC is between 50 % and 80 %, when net self-production is higher than zero. Changing the power outputs of the two-levels strategies don't provide any advantage to the final goal.

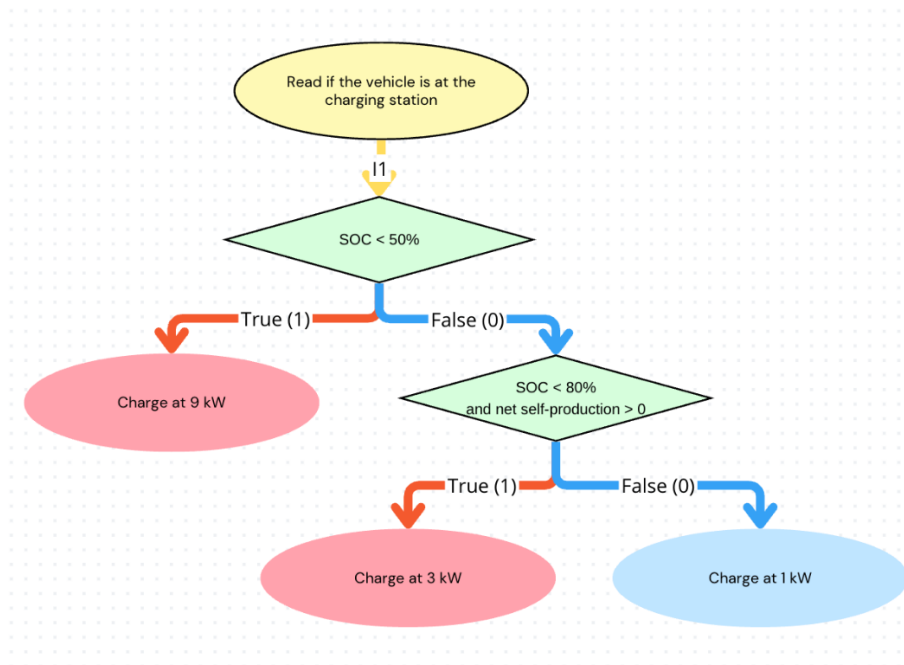


Figure 5: Control logic for the SC strategy for the EV

The results are shown in Table 1, compared to the baseline scenario.

Table 1: SC strategy results compared to Baseline scenario

Baseline_1001	SC implemented
---------------	----------------

Self-consumption fraction	22.9 %	31.9 %
From external grid [kWh]	7,455	6,291
CO ₂ emissions [kg]	3,999	3,375

As can be seen, the strategy concluded in positive results when considering only self-consumption indicators and environmental ones, with a 9% increase in SC, a higher increase than those found by R. Fachrizal and J. Munkhammar in similar studies [3]. Moreover, the PVs are able in this case to supply roughly 37 % of the EV charging demand on a yearly basis (dips of zero in December and peaks of 42 % in June) compared to the 16 % of the baseline. However, as seen in Table 2 the total system costs are higher, since when grid imports are required because not enough PVs production is available, the electricity prices are higher.

Table 2: SC strategy financial results compared to Baseline

	Baseline_1001	SC implemented
NPV difference (20 years) [SEK]	-	- 64,180
Energy supply costs Y1 [SEK]	20 282	23 879

3.1.2. Price Arbitrage

The second strategy to be implemented was PA, implemented on the Baseline 1001 model.

Firstly, two price ratio thresholds have been chosen: one of 0.3 (lower threshold) and a higher one of 0.75 (higher threshold). Two approaches of two (11 and 3 kW) and three (9, 3 and 1 kW) power output levels were implemented. Both the strategies result in a 15 % reduction in the energy supply cost, but between the last twos no significant difference is shown, meaning that for the cost minimization scenario, working with three different power outputs is not relevant. Moreover, to compare the results with other price ratio values, some sensitivity analysis was made, analysing different price thresholds while maintaining the same power outputs limits, but no relevant improvements were shown. The final controller logic is represented in Figure 6. The numerical results are represented in Table 3.

	Baseline_1001	PA implemented
NPV difference (20 years) [SEK]	-	+17 520
Energy supply costs Y1[SEK]	20 282	18 974

Table 3: Financial results of the PA strategy for the EV

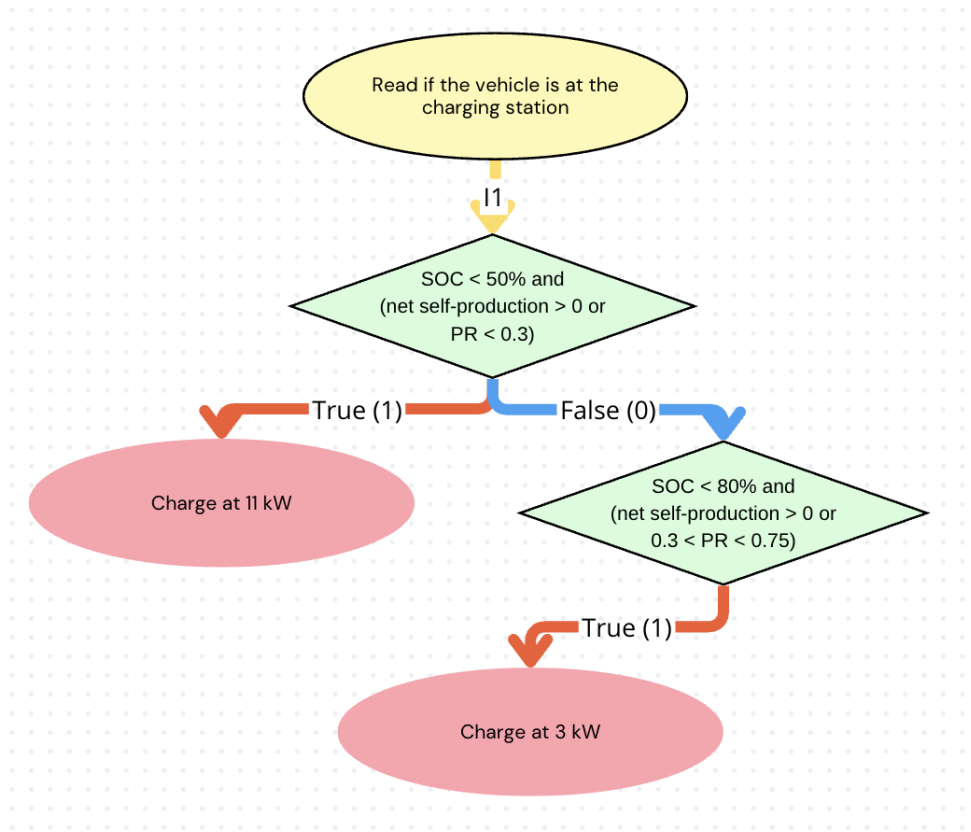


Figure 6: Controller logic for the PA strategy for the EV

3.1.3. Peak Shaving

Three different charging strategies were tried, with each of them utilizing equations which provided proportional power outputs. In this case, the main difference between the different tried strategies relies in the charging function that were used and are below listed.

$$\begin{aligned} \text{PowerOutput} &= (1 - \text{SOC}) * \text{ChargePower}, \\ \text{ChargePower} &= 4000 \text{ W} \end{aligned} \quad (3)$$

$$\begin{aligned} \text{PowerOutput} &= \text{ChargePower} * \left(\frac{1 - \text{SOC}}{1 - \text{SOC}_{\max}} \right) \\ \text{ChargePower} &= 800 \text{ W} \end{aligned} \quad (4)$$

$$\begin{aligned} \text{PowerOutput} &= \text{ChargePower} * (1 - \ln(1 + \text{SOC})) \\ \text{ChargePower} &= 3000 \text{ W} \end{aligned} \quad (5)$$

In all the three cases, the main pivot was the electricity consumption of the profiles value, used as a threshold to activate or not the EV's charging.

The second equation is a CC/CV charging strategy, so constant current – constant power. As long as the battery SOC is under a maximum value ($\text{SOC}_{\max}$ - set to 50 %), the EV is charged at a minimum power of 1400 W. Once the threshold SOC value is reached, the remaining 30 % is charged through (4).

The last strategy to be implemented is based on the logarithmic function (5), which results in the load profile shown in Figure 8.

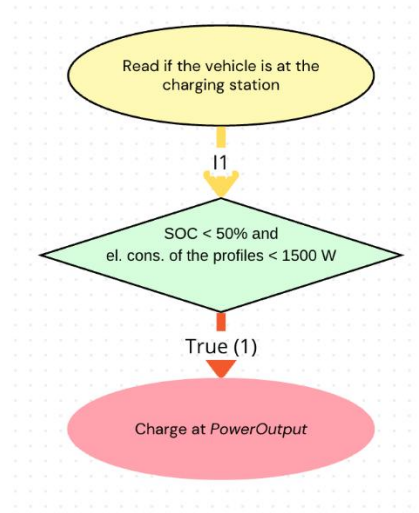


Figure 7: Control logic for the PS strategy for the EV

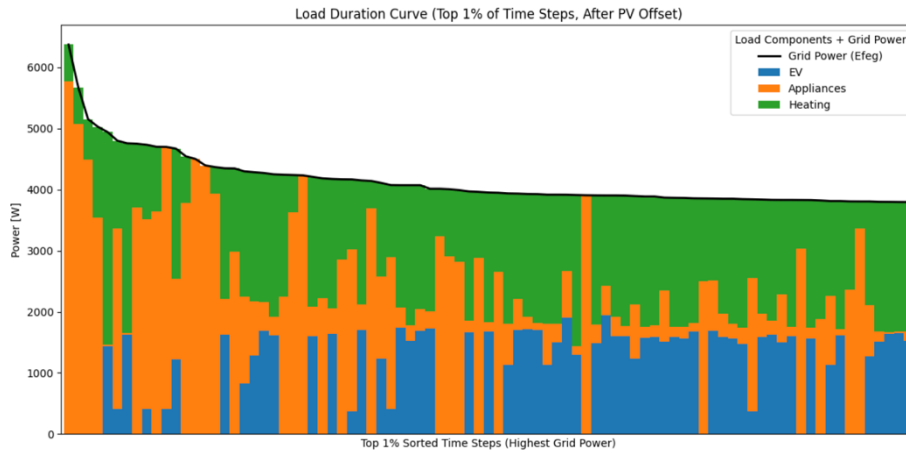


Figure 8: Top 1 % of load duration curve timesteps after PS of the EV

In all the three strategies, the previous peak of 7,408.0 W is reduced to 6,370.0 W, a hard limit imposed by a HP and appliances peak, which represents a 15 % reduction, in line with the range of 3%-20% peak reductions achieved by C. Loakimidis et al. in their research on EV fleet peak shaving [4]. Moreover, when looking at the two best performing strategies and checking their relative peak value of the EV, the logarithmic function provides a lower value of 1,958 W compared to the 2,353 W of the first strategy and to the baseline one of 3,041 W, so the strategy using the logarithmic function is selected as the optimal approach to EV PS.

3.2. Home Battery Operation Strategies

The E-level analysis for the battery is based on the baseline house 1012, supplemented with a programmable controller and a Sonnenbatterie Eco 4.5 system. This battery has a storage capacity of 4 kWh, a charging power of 4 kW, and a discharging power of 4.8 kW. The CAPEX of the battery is 49,000 SEK and the OPEX is 2,000 SEK.

3.2.1. Self consumption

Without the possibility of energy storage, the generated electricity must be consumed at the time of production to increase self-consumption. With an installed battery, dispatching self-produced electricity over time becomes possible. Surplus energy is used to charge the battery, while during periods of low PV generation the battery can be discharged to supply the load. This control logic is illustrated in Figure 9 and is applied to increase the household's self-consumption. The battery state of charge (SOC) is considered to respect the operational limits of the battery.

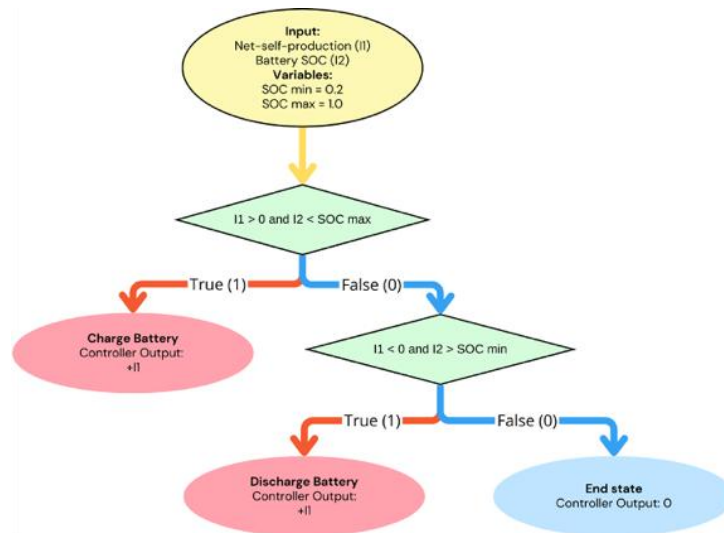


Figure 9: Logic self-consumption battery controller

The simulation results compared to the baseline case 1012 are shown in Table . The results for the main KPIs are positive. The self-consumption fraction increases by 9.6 % to 47.7 %, which lies within the typical improvement range of 20 – 50 % reported by Nyholm et al. [5]. The grid electricity imports decrease by 5.3 %, and CO₂ emissions are reduced by 4.7 %. However, the NPV shows a negative outcome due to the high initial investment costs.

Table 4: Results SC controller battery

	Baseline	SC implemented
Self-consumption fraction	38.1%	47.7 %
From external grid [kWh]	7,730	7,366
CO ₂ emissions [kg]	4,146	3,951
Battery annual discharge [kWh]	-	751
NPV difference (20 years) [SEK]	-	- 38,539
Energy supply costs Y1 [SEK]	20 760	19 788

3.2.2. Price Arbitrage

The price arbitrage strategy for the battery aims to charge the battery during periods of low electricity prices and discharge it when prices are high. Accordingly, low-price and high-price thresholds are implemented in the controller, as illustrated in Figure 10. The selected threshold

values (0.1 and 0.8) are based on the analysis presented in Section 2.3 and were further validated through a sensitivity analysis.

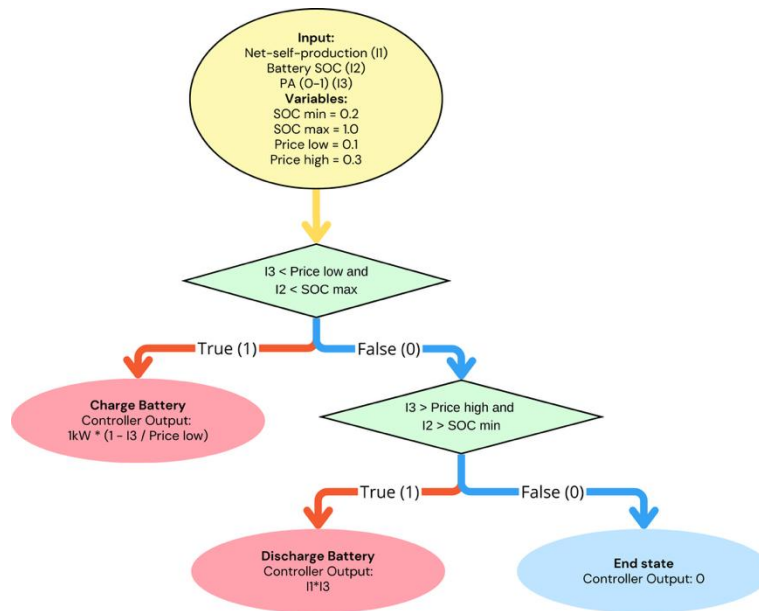


Figure 10: Logic PA controller battery

The results of the price arbitrage strategy, shown in Table , indicate that the controller operates as intended. Grid electricity consumption remains almost unchanged, with a slight reduction of 0.2 %, while the energy supply cost is reduced by 13.7 % compared to the baseline house. However, these savings are insufficient to make the battery economically viable, as the NPV remains negative. This finding is consistent with the study by Metz et al., which concluded that revenues from battery price arbitrage are generally insufficient to justify the investment and installation costs [6]. In addition, CO₂ emissions show a slight increase of 0.5 %.

Table 5: Results PA controller battery

	Baseline	PA implemented
Self-consumption fraction	38.1 %	40.2 %
From external grid [kWh]	7,730	7,764
CO ₂ emissions [kg]	4 146	4 165
Battery annual discharge [kWh]	-	361
NPV difference (20 years) [SEK]	-	-39945
Energy supply costs Y1 [SEK]	20 760	20 099

3.2.3. Peak Shaving

The logic of the peak shaving controller, shown in Figure 11, is to discharge as soon as the peak shaving threshold of 4.6 kW is reached to cover the difference between the threshold and the consumption. If the consumption is lower than 4.6 kW, the battery is charged with the delta between the threshold and consumption.

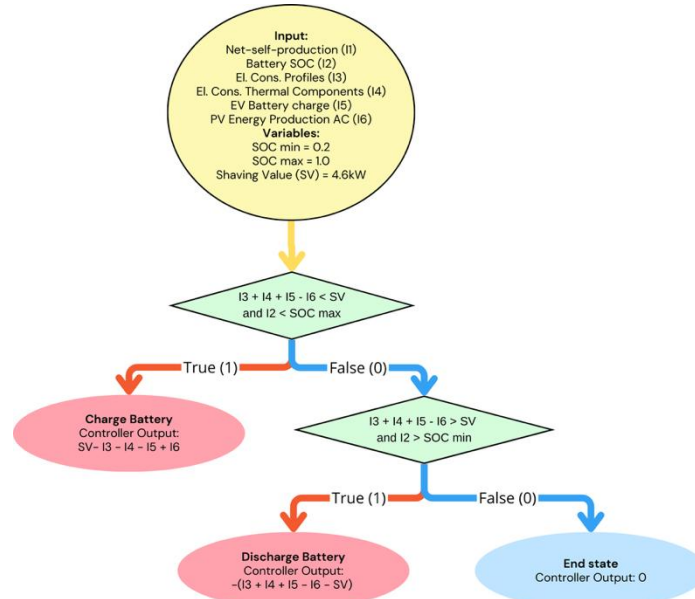


Figure 11: Logic PS controller battery

The duration curve in Figure 12 shows that the peak shaving strategy limits the peak grid power to 4.764 kW, which is 32.0 % lower than in the baseline case. A graphical analysis of this peak on 28 December at 10:00 (illustrated in Figure 13) shows that the peak is caused by a short period of high electricity demand from the household load profile, EV charging, and the heat pump operating simultaneously for approximately 10 minutes. During this period, battery recharging maintains a relatively high grid power demand, resulting in an average grid consumption above 4.6 kW.

The reviewed literature reports peak shaving reductions in the range of 42–49 % [7]. The lower reduction observed in this study is most likely due to the smaller battery capacity (4 kWh) and lower power rating compared to the literature values, which had a battery capacity of 22 kWh and power ratings of 5.2 kW installed.

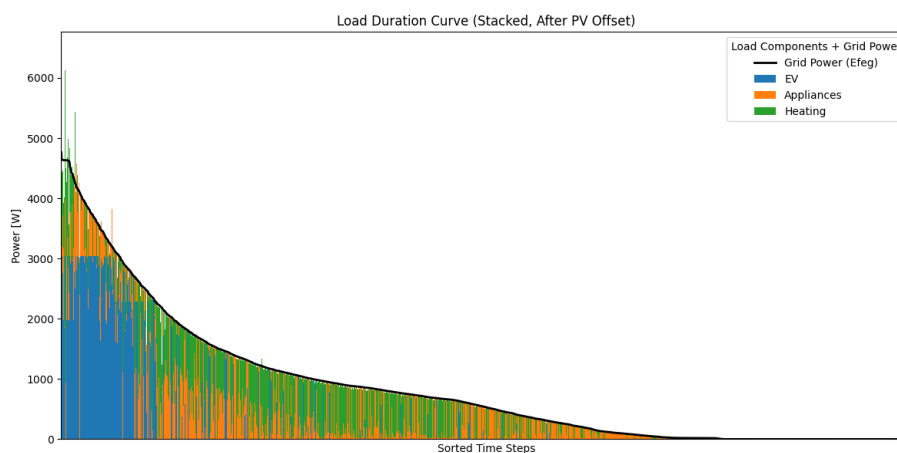


Figure 12: Load duration curve PS controller battery

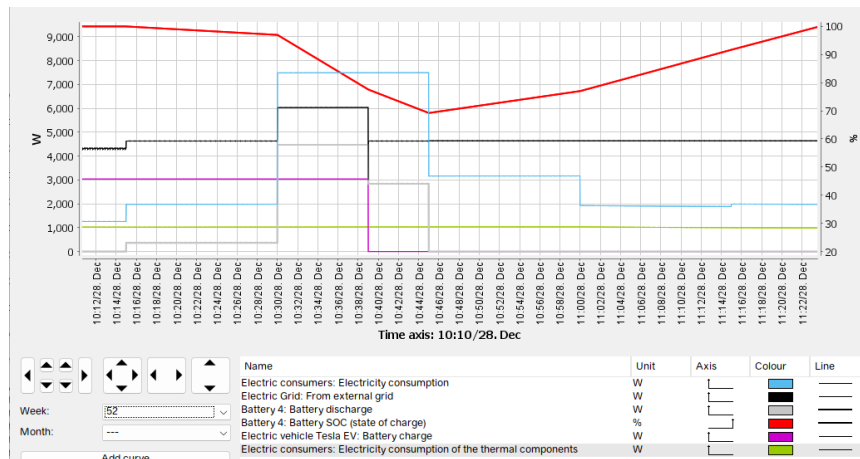


Figure 13: Max load peak from external grid

3.2.4. E-level Battery Strategies Results

The main KPIs and indicative factors of the simulation are visualised in Figure 14. The results show that all controllers behave as intended and each improves its targeted performance area. An analysis of the economic KPIs reveals that reducing the maximum peak load through the peak shaving controller has a higher economic impact than reducing electricity purchases via self-consumption or shifting consumption to low-price periods using price arbitrage. The PS controller achieves a reduction of 9.7 % in energy supply cost per unit of consumed energy. However, this reduction is still insufficient to make the battery investment economically viable, as the net present value (NPV) remains negative. These findings are consistent with the current situation in Sweden, having in mind that energy storage systems are subsidised by 50 % [8].

Regarding the environmental KPIs, the SC controller achieves the highest reduction in CO₂ emissions at 4.7 %, followed by the PS controller with a reduction of 3.2 %. In contrast, the PA controller shows a negative environmental outcome, with CO₂ emissions increasing by 57 kg (+0.3 %). The complete results are presented in Table 22 in Appendix VI.

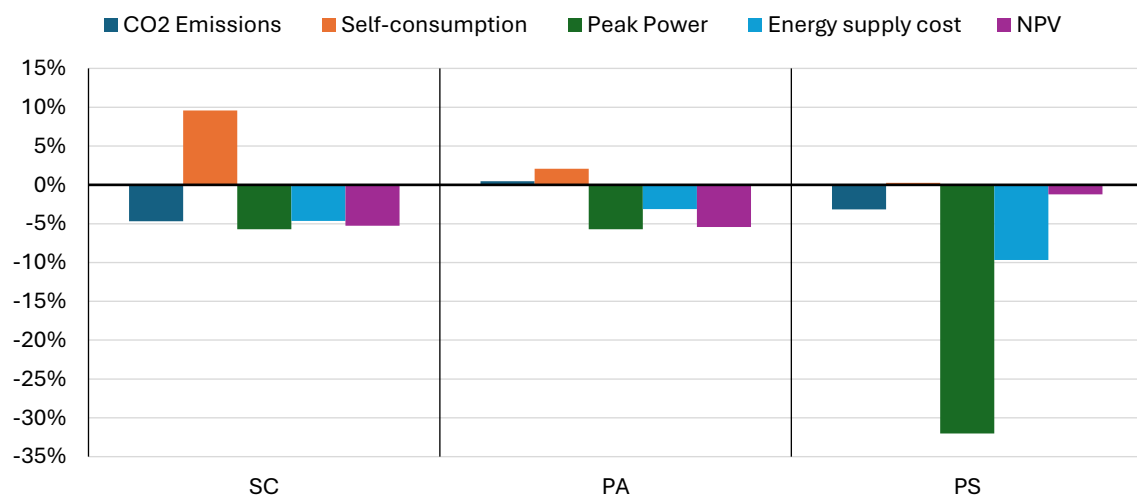


Figure 14: Relative improvements compared to baseline in % of different control strategy for battery.

3.3. Heat Pump Operation Strategies

The implementation of the three control strategies was applied to the heat pump (HP) and heating system for the property 1012, which uses electricity to heat a water tank for domestic hot water (DHW) usage, and provides space heating (SH) to warm up the building. Through the following smart control strategies, the operation of the HP is modified to maximize results across the 3 main objectives, whilst the number of switch-on times is tracked, which should stay below 2000 to preserve the health of the HP.

3.3.1. Self-consumption

The main approach to increase self-consumption with the HP involves shifting the load profile to maximize consumption at times when excess PV production occurs. Considering the two components present in the heat pump system, strategies were tested to modify DHW and SH operations in isolation and combined. The final strategy, labelled SC(DHW+SH) combines both DHW and SH control. Whilst different thresholds for net self-production were explored, a 0 W threshold was found to perform adequately for the studied house.

During periods of excess PV production, the DHW dT threshold, which is the desired difference in temperature of the DHW tank over the setpoint temperature, is raised to 10 K, and otherwise left at the 5K difference of the baseline. Regarding the SH, a heating rate threshold is introduced, which is 100 K when there is a positive net PV production, and otherwise it is – 60 K. The rate modifies the heating curve, by changing the threshold of degree minutes above which the HP runs at minimum power. Furthermore, it modifies the condition to turn on SH, by changing the thresholds of degree minutes to the new heating rate variable, such that the HP provides less SH during periods of no excess PV production. The logic of the HP SC control is shown in Figure 15 below.

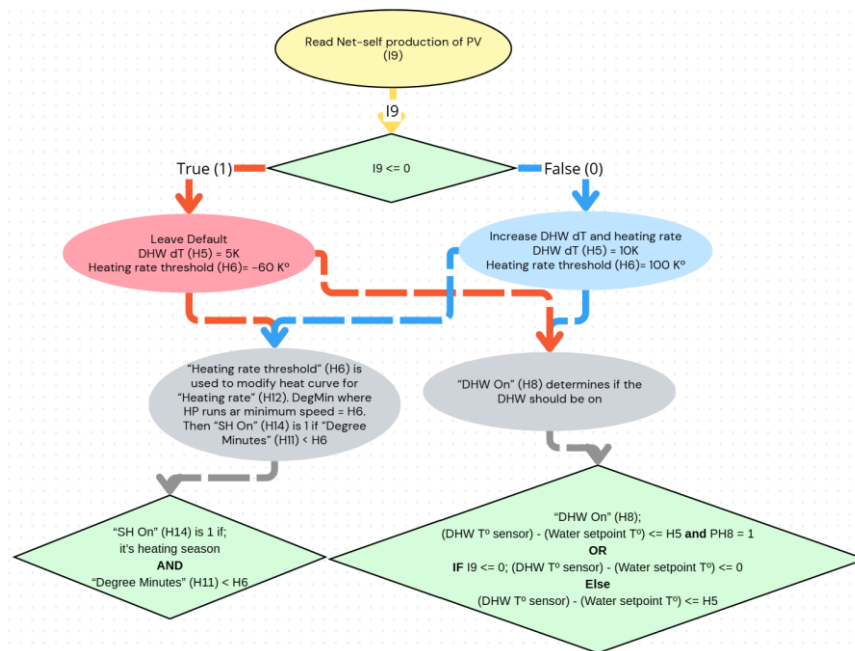


Figure 15: Logic diagram of the heat pump SC strategy

3.3.2. Price arbitrage

The price arbitrage strategy operates by pre-charging the building or the DHW tank at higher temperatures than demanded while electricity prices are low, by shifting the heating curve (for space heating) and DHW dT by a specified temperature difference, depending on how high the relative prices are. The *DHW dT* threshold is modified as in the SC strategy, and the heating curve is modified to affect SH, by adding a *Heat Curve Mod dT* which is added to the final calculated heating curve of the original HP control. This variable then affects the degree minutes and the heating rate of the HP; If it is positive, the heating rate is larger, and the building will be warmed up more than demanded. If it is negative, the heating rate is also smaller, and the building will get colder than demanded. The logic for changing the variables according to relative prices was implemented according to Figure 16 shown below.

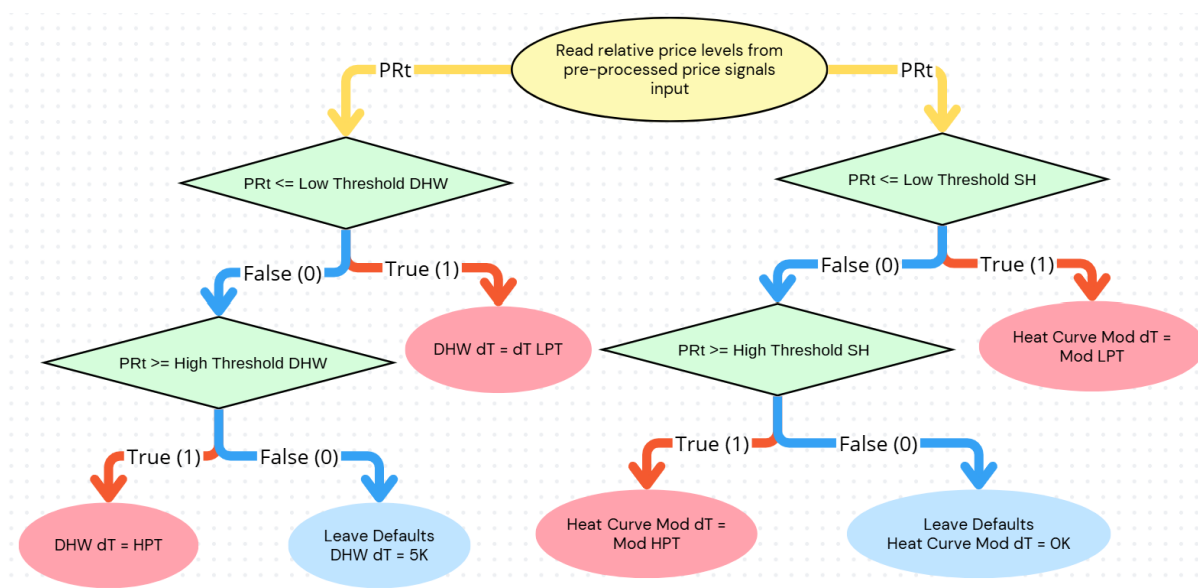


Figure 16: Logic diagram of the heat pump PA control strategy

As in SC several strategies were attempted, including the control of either DHW or SH, as well as a combined, which yielded optimal results. Thresholds are 0 for the Low Price Threshold (LPT), 0.5 and 0.85 for the High Price Threshold (HPT) of the DHW and SH, respectively. The LPT is specifically chosen dT (both for DHW and HC) = 0, because especially for space heating, its heat losses are very small, and the temperature change is not as large. Therefore, it is enough to pre-charge the building while relative price = 0 in 24 hours and the next time to do this only when the relative price drops back to 0. This strategy is only working well because the prices have a period of high and low prices of around 24 hours, and the building cooldown also takes around 24 hours. This long delay in heat response is possible due to the floor heating utilized in the studied homes and would not be as effective when using air's thermal inertia as a means of thermal storage. The DHW dT is 6 K for LPT and 4 K for HPT, whereas the *Heat Curve Mod dT* is 12 K and -9 K for LPT and HPT respectively.

3.3.3. Peak shaving

To reduce the peak power surcharge of the final electricity bill, peak shaving was explored as a strategy to reduce HP consumption during periods of high consumption from the appliances and EV. Two main strategies are considered, namely smoothing the HP's own power peaks, and turning off the HP during periods of high consumption from appliances and EV. The first approach does not significantly reduce peak power, as the overall power import peaks do not align with the individual HP peaks being reduced. A much more effective strategy consists in turning off the heat pump during peak power periods, defined through a threshold for the sum of EV and appliances consumption minus PV production. This control logic is found in Figure 17 below.

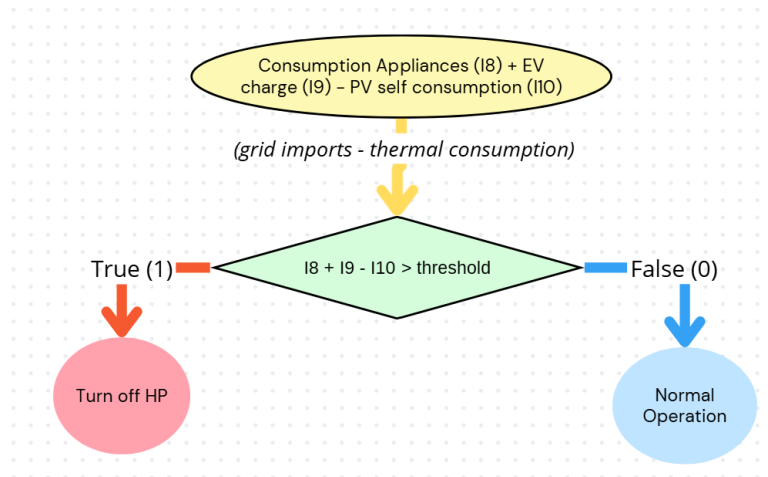


Figure 17: Logic diagram of the heat pump PS control strategy

To determine the optimal threshold of EV and appliances peak P under which the HP is turned off, values between 2 kW and 4 kW are explored for the reference 1012 house. As shown in the load duration curve of Figure 18 below, a 2.8 kW threshold achieves highest possible peak power reduction, given that the new peak is composed of EV and appliances consumption and cannot be reduced through any HP control strategies.

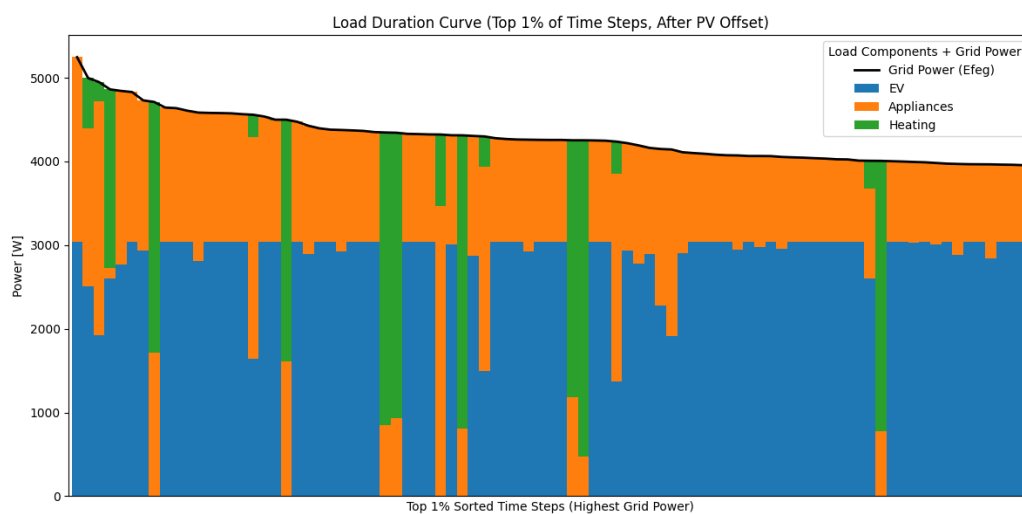


Figure 18: Load duration curve with a 2.8 kW peak power threshold

3.3.4. E-level Water Storage Tank Size Sensitivity

To test the sensitivity of results to the hot water tank size, the best performing strategies were explored, keeping the current logic constant, and changing the tank size between 200 and 600 litres, adding 500 SEK in investment costs for each 100-litre increase in tank size. Appendix III presents the results of the sensitivity analysis for house 1012, where it was determined that a 200l tank remains the optimal solution for PA, given the low influence of tank size on this control logic. For SC, a larger DHW tank increases the amount of excess PV that can be dumped into the DHW tank, and for PS, it reduces the impact on occupant comfort of switching off the HP and enables greater costs reductions through more aggressive peak power thresholds. The best trade-off between tank costs increases objective improvements is for a 400l tank. In both SC and PS, increasing the tank size beyond 400 litres does not significantly improve the KPIs, whereas it represents a larger investment cost, which is not economically sensible. Similar sensitivity analyses were applied to all households, determining the optimal tank size based on house specific consumption profiles, which is 500 litres for houses 1005 and 1008, and 200 litres for all others.

3.3.5. E-level Heat Pump Control Strategies Results

Table 6 below shows the main results of the optimal HP control strategies, and Figure 19 shows the relative increase across the three main indicators of each approach.

Table 6: Key metric results for the three control strategies involving the heat pump

	Baseline	SC – 400l tank	PA – 200l tank	PS – 400l tank
Energy Deficit [kWh]	56.7	45.9	58.2	58.4
Building Min. T [°C]	20.6	20.5	20	20.5
DHW tank Min. T Top Layer [°C]	48.7	50.5	47.5	47.8
Comfort change Vs. baseline [%]	-	+ 4.49%	-13.14%	-5.33%
Self-consumption fraction [%]	38.1	41.8	37.7	35.3
CO2 emissions [kg]	4,146	4,053	3,902	4,159
Electricity consumption [kWh]	8,900	9,023	8,408	8,687
From external grid [kWh]	7,730	7,556	7,275	7,544
Energy supply costs Y1 [SEK]	20,760	20,175	18,978	18,862
NPV difference compared to reference system [SEK]	-	+ 5,758	+26,524	+27,904
Peak power import [W]	7,007	6,609	6,425	5,246
Average cost of electricity [SEK/kWh]	2.33	2.24	2.26	2.17

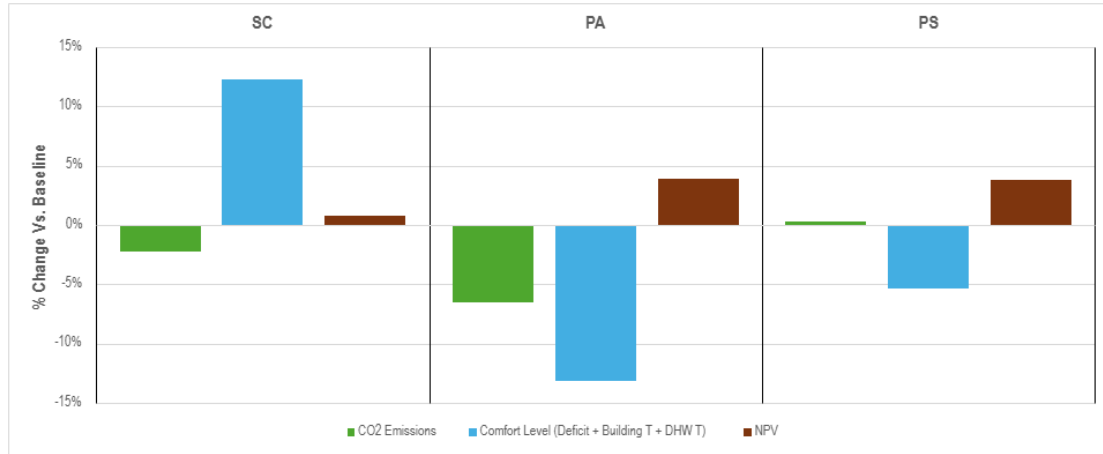


Figure 19: Relative improvements compared to baseline of key performance across the three main objectives for HP control strategies.

Using the HP for self-consumption is a solid strategy, achieving 41.8 % of SC, which aligns with similar studies [9] that achieved a 37 % PV self-consumption through smart HP control. Price arbitrage is also an effective strategy, reducing energy supply costs by 8.5 %, but this is largely driven by a reduction in energy consumption, given that only a 3 % reduction is achieved in the average costs per kWh. Whilst within the range of improvement achieved by similar papers[9], [10], the PA strategy is significantly lower than the best performing rules-based strategies, which can achieve costs reductions larger than 20 %[11]. This is likely due to the relatively low price-spread observed in the electricity prices, which not only reduces potential profits, but also leads to a higher frequency of activation of the PA control actions. Due to the inflexible nature of heat pumps, deviation from ideal operation in each time-step will lead to greater power consumption to meet the required heating demand, such that the strategy would be more effective if used in moderation, for time-steps with extreme prices.

Peak shaving is the most effective strategy with the heat pump and can reduce peak power consumption by 25 %. Whilst significant, the reduction is lower than the 38 % reduction achieved by J. Clauss et al. [12] for their HP control strategy in Norwegian residential buildings. The discrepancy is likely due to a lower initial peak in our scenario, given that the final peak stems from appliances and EV and isn't remediable through HP control.

The strongest strategy is different for each objective considered. Whilst occupant comfort levels are within an acceptable range for all scenarios, they only improve for the SC strategy. Regarding the environmental impact, PA is the most effective strategy, whereas SC only reduces emissions slightly, and PS causes a moderate increase caused by a lower self-consumption fraction. Finally, HP control for peak shaving yields the lowest TLCC, followed closely by price arbitrage, with self-consumption having the weakest effect. However, this is in large driven by the lower total consumption of the first strategies, particularly price arbitrage, where total energy consumption is 5 % lower than in the base case. Analysing the average cost of electricity per kWh consumed provides complementary insights into each strategies' effectiveness and suggests a better performance of SC over PA in optimally managing the available energy. Ultimately, there exist trade-offs in objective completion for all approaches, so a combination of all three strategies that compensates individual shortcomings will be the most effective smart control solution.

4. C-level analysis: Combined Strategies and application on multiple buildings

To develop the best controllers for each technology, SC, PA and PS strategies were combined at an individual technology level, and subsequently stacked on top of other technologies, to develop the best possible smart control system to manage the energy consumption of decentralized homes.

4.1. Batteries Combined Strategies

For the implementation of the different control strategies into one combined one. The PS-controller is the base controller for it due to the greatest improvement in the KPIs (environmental and economic). As shown in the Figure 20, the discharging is implemented in the same way then the PS-controller, discharging when the demand is higher than the threshold (4.6 kW). For the charging, the logic of PS and SC was implemented first to prioritize the self-consumption over electricity from the grid. To not charge from the grid when the electricity is high, the PA threshold is implemented.

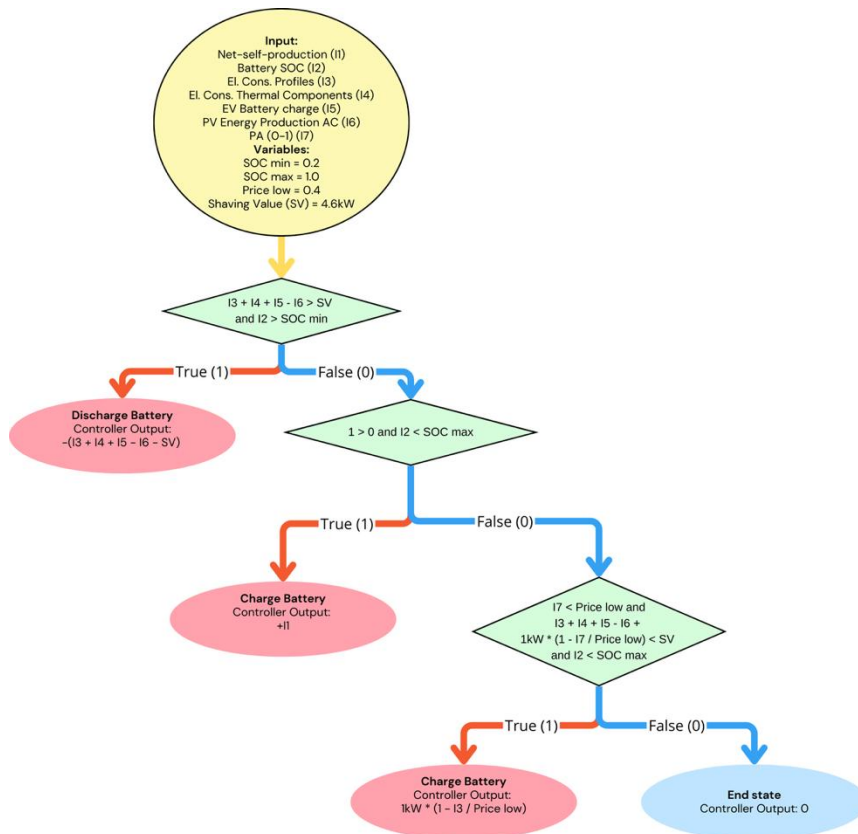


Figure 20: Logic of Combined controller battery

The results of the simulation show that the combined controller provides a slight improvement compared to the PS controller across the main KPIs visualised in Figure 21. Annual CO₂ emissions are reduced by 3.6 % to 3,997 kg. The peak power consumption from the grid decreases to 4,754 W, which is 32.2 % lower than in the base case. In addition, the energy supply cost per kWh is reduced by 10.2 %, representing the lowest value among all battery controller

strategies. The NPV difference is also the smallest at –7,576 SEK but nevertheless remains negative. Including the Swedish battery subsidies of 50 % a battery is economical viable with a positive NPV of 16,924 SEK. The complete results are presented in Table 22 in Appendix VI.

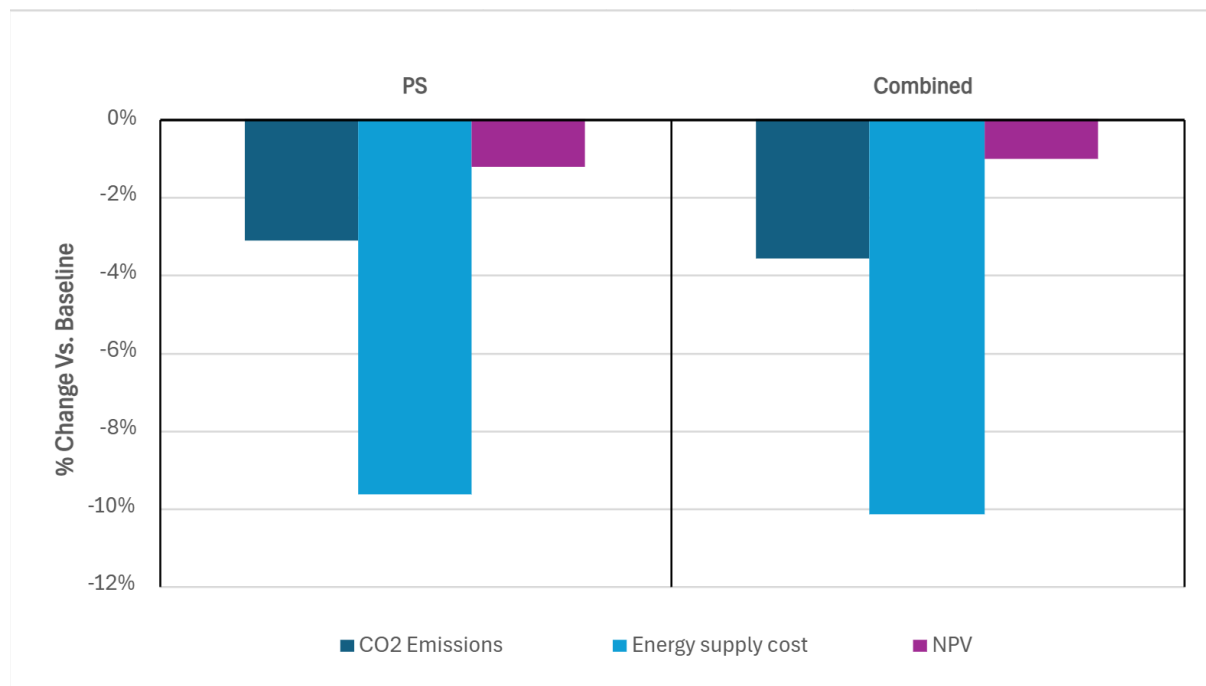


Figure 21: Relative improvements comparison to baseline in % of KPI for PS and Combined controllers.

A sensitivity analysis for battery sizes between 2 kWh and 16 kWh was conducted in Polysun, but no configuration yielded better economic results than the 4 kWh battery. The battery was also implemented in houses 1001 – 1011, yet the NPV difference remained negative in all cases. Therefore, without subsidies, the battery system is not economically viable.

For this reason, the battery is not further considered in the optimisation of the integrated control strategy involving the heat pump, EV, and battery.

4.2. Electric Vehicle Combined Strategies

The EV's C-level strategy aimed to combine the earlier developed standing alone self-consumption, price arbitrage and peak shaving schemes, now focusing on having a single controller which provides better values of self-consumption fraction, NPV and energy supply costs and peaks than the baseline at the same time. The results only look at EV's performances.

Figure 22 shows the logic behind the controller.

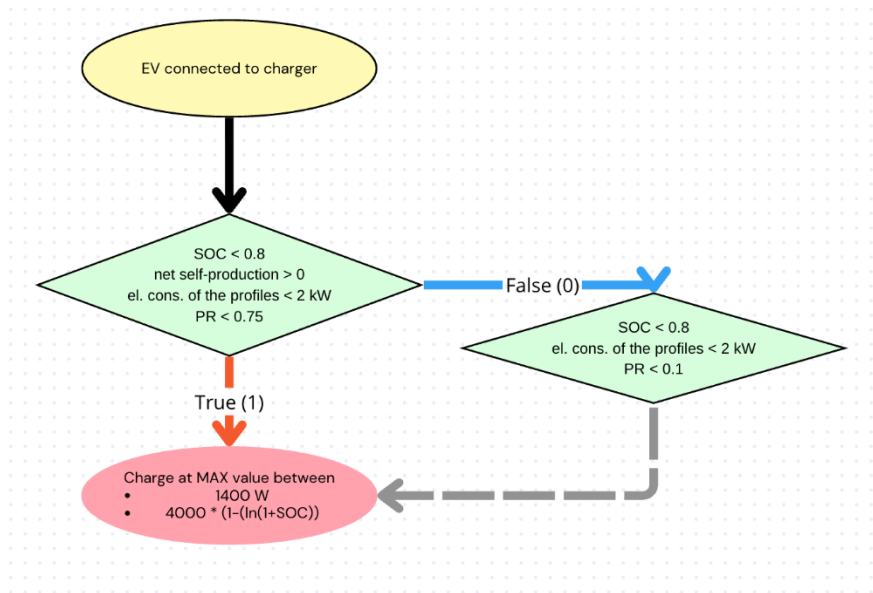


Figure 22: Control logic behind the EV controller

The results are instead presented in Table 7.

Table 7: Results C-Level controller EV

	Baseline	C-level controller
Self-consumption fraction	38.1 %	45.5 %
NPV difference (20 years) [SEK]	-	1,044
Energy supply costs (20 years) [SEK]	504,421	499,768
Peak power import [W]	7,007	6,389
CO ₂ emissions [kg]	4,146	4,762

As can be seen, the results are better in every aspect when looking at the economics, self-consumption indicator and peaks reduction. However, a rise in the CO₂ emissions happens, most probably because of the 14 % increase in kWh imported from the grid.

4.3. Heat Pump Combined Strategies

To combine the multiple individual strategies, the best performing SC, PA and PS were aggregated into a single HP controller. Owing to the logic of control strategies, peak shaving takes preference, turning off the HP during periods of peak consumption. When aggregated consumption is below the peak, power consumption is increased during periods of excess PV production, and finally the PA logic is implemented, modifying the normal operation when relative prices are above or below the thresholds defined in the previous section. The logic diagram for the combined HP controller is illustrated in Figure 23.

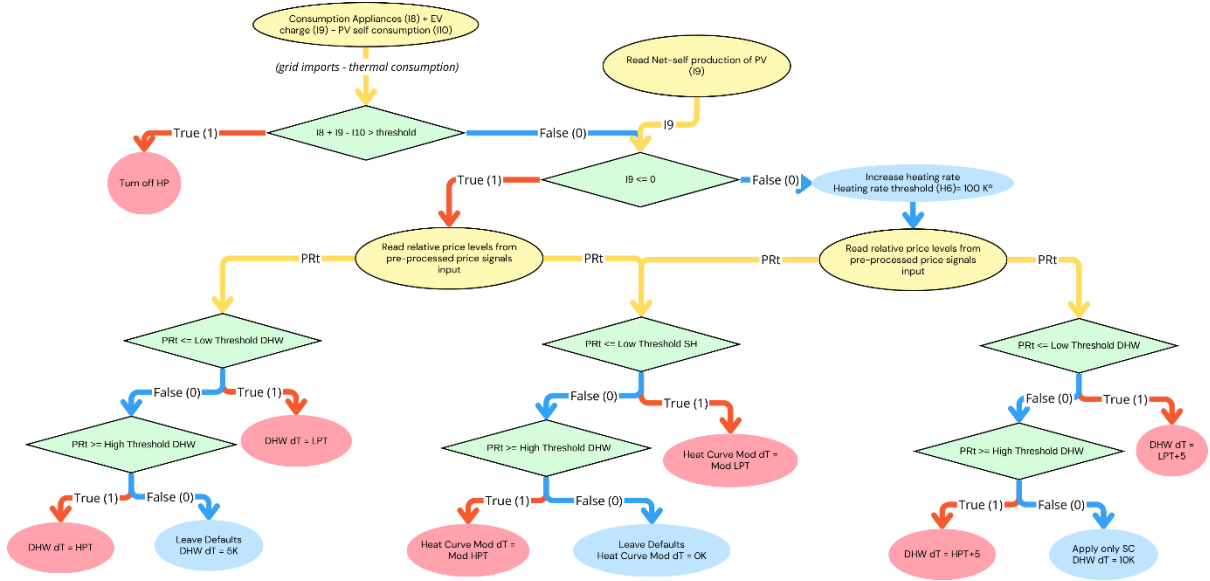


Figure 23: Logic diagram of the combined heat pump control strategy

By combining the three approaches, the increased comfort level of SC offsets the negative impact of PS and PA, which themselves have a far greater impact in costs reduction. Cascading all three logic controls yields a greater improvement over the baseline than combining pairs of control strategies, as shown in Appendix IV. Ultimately, the combined control strategy achieves a 4.8 % reduction in CO₂ emissions, an 11.59 % reduction in average costs of electricity, and an increase in comfort of 15 %, whilst maintaining the number of switch-on times for the heat pump at 1,371 per year.

4.4. Multi-technology control strategy

The multi control strategy of our system comprehends both EV and HP C-level controllers. Only an adjustment was done. Now the charging logarithmic function of the EV is multiplied times two. On the other hand, the battery is excluded because of its economic unfeasibility without subsidies, while EV + HP work with good performance together. In fact, building the vehicle combined controller on top of the HP one, provides better results than only regulating HP operation. Table 8 summarizes the results, considering several indicators.

As can be seen, the results show better values in every aspect compared to the baseline scenario, which mean the strategy correctly works and is preferable than the standard case. However, some analysis should be done to understand if the model is also superior to the one with the only HP controller implemented, which by itself it is more performative than the only EV one. Doing so, it can be noted that indicators like the self-consumption or electricity consumption are respectively lower than the only HP and even lower than the baseline model. However, when looking at the environmental impact of the system (CO₂ emissions) and especially at the economics, the results are promising. In fact, while the reduction in emission is of a roughly 9 % compared to the only HP's 5 %, when looking at the economic results, particularly at the energy supply cost and the NPV, they show a reduction in energy costs of 18 % compared to the 13 % of HP only and an increase of NPV (decrease in TLCC) 47 % higher than only HP control.

Table 8: Key metric results for HP, EV and HP+EV combined control strategies

	Baseline	HP Combined	EV Combined	HP+EV Combined
Energy Deficit [kWh]	56.7	46.6	56.8	47.4
Building Min. T [°C]	20.6	20.7	20.5	20.2
DHW tank Min. T Top Layer [°C]	48.7	47.1	47.1	46.9
Comfort change vs. baseline [%]	-	+15.02%	-3.95%	+10.76%
Self-consumption fraction [%]	38.1	40.9	45.5	39.2
CO2 emissions [kg]	4,146	3,945	4,762	3,762
Electricity consumption [kWh]	8,900	8,753	8,895	9,813
Energy supply costs in Y1 [SEK]	20,760	18,027	20,568	16,980
NPV difference compared to reference system [SEK]	-	+37,923	+1,044	+55,569
Peak power import [W]	7,007	5,246	6,389	5,560
Average cost of electricity [SEK/kWh]	2.33	2.06	2.31	1.73

4.5. C-level control for 12 buildings

When applying the combined HP + EV controller to each house, some minor adjustment in the control logic were made both for the EV and for the HP controllers.

First, the HP controller was implemented to every building. For the part controlling the SH, nothing needed to change, as the SH is only relevant for the PA and the electricity prices stay the same for every building. We kept also the PS threshold as the 2,8-kW limit seemed to fit all the houses well. The change happened mainly for the DHW thresholds and for the size of the water tank. The thresholds were changed to keep the maximum switch on-off times of the HP under 2,000 a year and the tank size was varied to maximize the storage capacity, if the house uses a lot of hot water – therefore for the Buildings 5 and 8, changing the tank size decreased the energy costs sufficiently to balance for the additional tank capacity costs. Similarly, as the houses 1001, 1002, 1004, 1006, 1007 and 1009 have comparatively lower hot water consumption, the price change wasn't as significant and to reduce the switching times below 2,000, it was needed to turn off PA for the DHW dT.

The detailed control logic that was changed in the HP control for every household is given in the Appendix VII.

For the EV, the maximum charging rate was increased from 4 kW to 8 kW. This change was based on some sensitivity analysis. It is needed to mention, that with the additional EV control, the HP

switch on-off times increase and for the houses 1003 and 1010 the switch on-off times rise slightly above 2,000.

In this section the results of the single adapted control strategies for each house are summed (or the average is taken depending on the KPI) and compared to the baseline total. The results are shown in Table 9.

Table 9: Neighbourhood comparison of baseline buildings - total to smart HP and EV controllers - total

	Baseline - Total	HP+EV Combined - Total
Energy Deficit [kWh]	637.4	486.6
Building Min. T [°C]	20.4	20.0
DHW tank Min. T Top Layer [°C]	48.4	46.9
Comfort change Vs. baseline [%]	-	+18.6%
Self-consumption fraction [%]	25.9	26.8
CO2 emissions [kg]	39,388	34,642
Electricity consumption [kWh]	84,353	82,450
Energy supply costs in Y1 [SEK]	220,239	168,672
NPV difference compared to reference system [SEK]	-	730,096
Peak power import (Billed) [W]	89,346	62,625
Average cost of electricity [SEK/kWh]	2.61	2.04

The results presented above are encouraging and show an overall increase in the economic, environmental and comfort metrics. A 24% reduction in energy deficit is reached, with an 18.6% increase in comfort, and a 12% reduction in CO₂ emissions. The supply costs are reduced by 23.4% during the 20 years, together with an NPV increase of 9%.

4.6. Heterogeneity of Benefits and Impact on Homeowner Satisfaction

An analysis of objective accomplishments for smart control in individual homes, as shown in Table 9, reveals a high degree of heterogeneity in the benefit obtained by households. These discrepancies arise due to differences in energy deficit, indoor comfort parameters, self-consumption, energy consumption, peak import, and emissions from individual homes, which are a result of different characteristics affecting the relative impact of smart control strategies. Some homes will own extremely economical systems with enhanced KPIs, whereas others would experience a limited benefit in economic terms as well as a slight comfort penalty. This heterogeneity in benefits, to a certain extent, could be a driving factor for dissatisfaction among different members of a community, as members of certain households who feel that their smart controls do not yield the same benefits as other homes would consider it an unfair system. However, all homes see an improvement over no smart control implementation, and as such it is likely that widespread adoption of smart controls would take place.

5. A-level analysis: Integrated community control strategies

For the A-level, an Energy Community (EC) development was analysed for the collection of 12 houses in the simulated neighbourhood. In its push to promote the development of such solutions as a companion of the green energy transition, the EU has stated its ambitions to see developments of energy communities, defined as “the collaboration of local stakeholders who produce, consume, and manage their own energy for the primary purpose of creating economic, environmental, and social benefits for the community” [13].

The proposed EC shifts the point of grid connection to the neighbourhood’s substation, changing the peak power import measure point, and reducing the peak power surcharge by replacing the individual household peaks by a community averaged value, charged at the same peak power price. Furthermore, the community increases self-consumption through the sharing of excess PV production between different households in the community. Unlike in a collection of decentralized homes, the energy community provides district heating through a centralized ground-source heat pump, which stores water in two tanks, each of 4000 litres, for domestic hot water and space heating respectively. Whilst the twelve EVs remain independently controlled, a centralized BESS is analysed. The costs of new components for the community system are highlighted in Table 10 below.

Table 10: Financial inputs for community level system components.

Cost included	Unit price	O & M Costs	Service life [years]
Heat pump	3,200,000 SEK	20,000 SEK/y	30
Community microgrid	800,000 SEK	-	100
Community battery	60,000 SEK +3,000 SEK/kWh	4,000 SEK/y	Automatic calculation

By aggregating residential users under one energy community, with the results shown in Table 12 for the Baseline A-level, significant savings can be achieved over a collection of independent homes. Despite additional investment costs of 1.6 MSEK, the annual electricity supply costs are reduced by 31.3%, TLCC are 8.9% lower, and CO₂ emissions are 15.5% lower. Whilst the comfort level is reduced by 3.78%, this is mainly driven by an increase in the energy deficit, whilst the minimum building and DHW tank temperatures are higher than in the base case.

The improved performance is owed in large to the 35.6% higher self-consumption through PV sharing, and the 23.3% reduction in peak power imports. With these benefits, an uncontrolled energy community already presents greater improvement on the three objectives than implementing smart controllers in a collection of independent houses, as discussed in the C-level analysis. To improve the energy community, smart controls are explored for all technologies, to devise the best strategy for communal energy management.

5.1. Community level battery control strategies

For the A-level controller, the same combined strategy as used for the C-level is implemented, with several adjustments to the input data; The EV loads were aggregated, PV production was summed up.

A sensitivity analysis was conducted to determine suitable values for battery capacity, peak shaving, and price arbitrage thresholds. The selected parameters were a low price arbitrage threshold of 0.3, a high price arbitrage threshold of 0.7, and a peak shaving limit of 30 kW. Three batteries are analyzed, sized at 10, 20 and 30 kWh, rated at 0.9, 0.45 and 0.3C respectively configurations were analysed: Pro AC9-10, Pro AC9-20, and Pro AC9-30, all with a charge and discharge power of 9 kW and storage capacities ranging from 10 to 30 kWh. The economic evaluation in Figure 24 shows that the NPV is lowest for the larger battery capacities, with 20 kWh and 30 kWh batteries achieving peak shaving reductions of 18% and 19.9%, CO₂ emission reductions of 0.2% in both cases, and energy supply cost reductions of 5% and 5.5%, respectively. However, the additional investment required for the 30 kWh battery

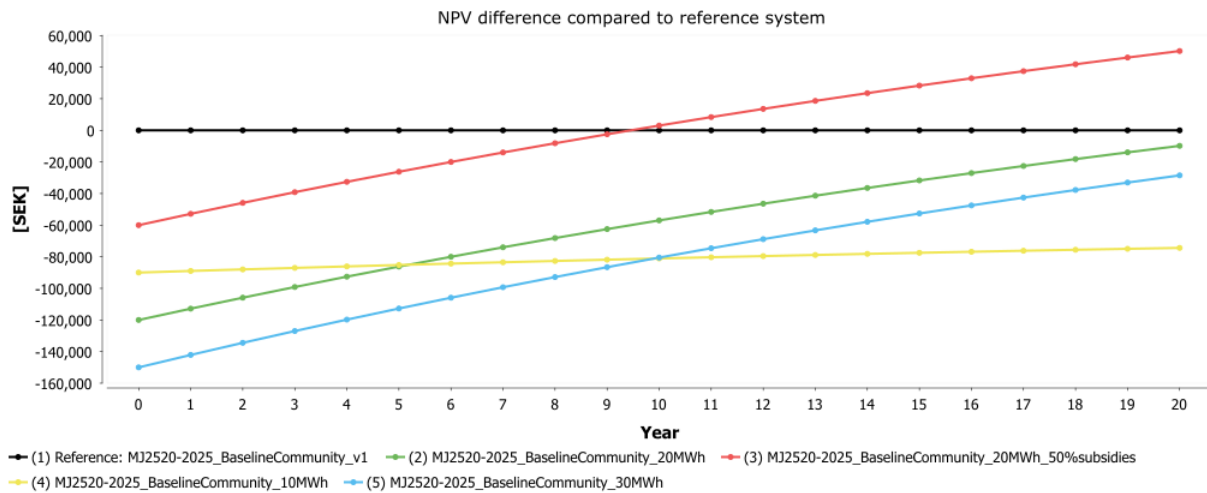


Figure 24: NPV difference 10-30MWh battery

is not economically justified, as the difference in NPV is almost 20 kSEK. The 10 kWh battery is too small to provide effective peak shaving and is therefore not competitive with the other two configurations. All the results of the different parameters are shown in Table 11.

When considering a subsidy of 50 % [5], the system becomes profitable in the tenth year, resulting in a positive NPV of 50,182 SEK (+0.7 %). Since this project assumes no subsidies, the battery is not considered further.

Table 11: Result comparison SC, PA, PS & combined control strategy battery (relative improvements compared to baseline in %)

	Baseline-A-Level	Combined 10kW	Combined 20kW	Combined 30kW
Energy Deficit [kWh %]	829	829 0.0	829 0.0	829 0.0
CO2 emissions [kg %]	33,269	33 322 -0.2	33 328 -0.2	33 336 -0.2
Battery discharging energy [kWh %]	-	265	292	296
Electricity consumption (Ecs) [kWh %]	82333	82336 0.0	82336 0.0	82336 0.0
Peak power import [W %]	41122	40051 -2.6	33721 -18.0	32951 -19.9
Self-consumption fraction [% %]	35.1	35.2 +0.1	35.2 +0.1	35.2 +0.1
Energy supply costs Y1 [SEK %]		150 150 -0.7	143 689 -5.0	142 916 -5.5
NPV difference compared to reference system [SEK %]	-	-74399 -1.0	-9818 -0.1	- 28499 -0.4
Average cost of electricity [SEK energy supply/kWh consumed %]	1.84	1.82 -0.7	1.75 -5.0	1.74 -5.5

5.2. Community level EV control strategies

The community level controller for the battery was kept the same as for the C-level, except for the upper limit of 22 kW for the electricity consumptions of the profiles that has been implemented to correctly charge all the vehicles.

As can be seen in Table 12 below, better results are achieved comparing to the decentralized smart neighbourhood, especially when looking at the peak imports and apart of the CO₂ emissions. However, the implementation of decentralized smart controls for EV charging leads to a higher peak power, stemming from uncoordinated increases of EV charging power across all residences, which negatively affect the supply costs for the community. Consequently, it is not advisable to implement decentralized EV controls, and instead a new centralized solution should be developed, or alternatively the default EV charging schedule should be maintained.

5.3. Community level heat pump control strategies

As opposed to the air-source HP present in the individual households, the energy community relies on a centralized ground-source heat pump, to charge two storage tanks for DHW and SH, which are 66% larger than in a decentralized scenario.

The SC controls modify the operation, but beyond increasing consumption under excess PV production, the larger thermal storage enables a reduction in heating power under conventional operation, further increasing the strategy's profit potential. The cut in temperature of the DHW tank's top layer, which defines the level at which the HP is activated to supply heat for DHW, is increased by 3 °C to 63 °C, whereas it is reduced by 2 °C under normal operating conditions. The cut-out temperature, which indicates the setpoint DHW tank temperature at which the HP is turned off, is reduced by 2 °C to 63 °C under normal operating conditions, but it is not increased under excess PV conditions, to avoid breaching the tank's maximum T° limit. The SH is controlled by adding 4 °C to the final heating curve, increasing the setpoint temperature under excess PV. To avoid excessive flickering on and off, an 800 W threshold is defined for net-self production.

The PS strategy follows the same logic implemented in the individual HP control, turning off the HP for peaks of EV and appliances net consumption above 19 kW (that is subtracting PV self-consumption). Finally, PA is implemented only to the SH tank, by increasing the final heating curve by 3 °C under relative prices below 0.6 and decreasing it by 8 °C for relative prices above 0.98, whilst decreasing it by 1 °C under normal operation.

Similarly to the individual HP control, PS takes priority over the HP's control, followed by SC and ultimately PA, as illustrated in Figure 25 below.

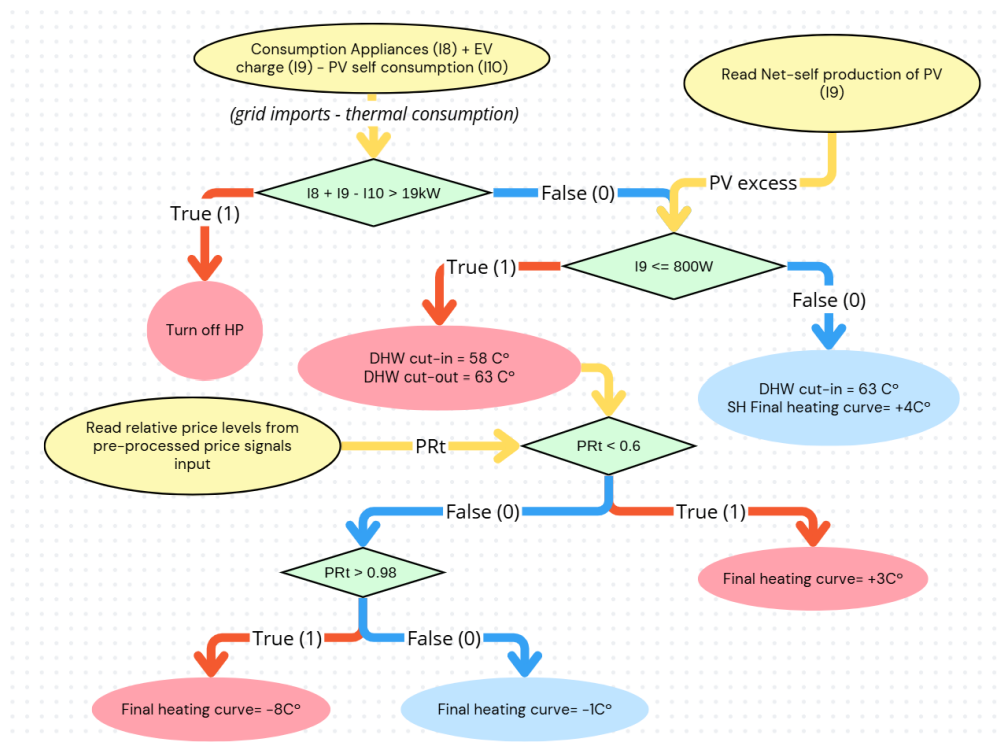


Figure 25: Logic diagram of the centralized heat pump control strategy.

5.4. Smart Energy Community Results

Comparing the results of implementing smart technology controls in the energy community, Table , we can observe that HP control yields the best results, whereas EV and BESS (without subsidies) yield higher TLCC. Stacking controller technologies yielded similar challenges, due to unwarranted interactions between control logics, and the higher CAPEX of BESS, for which a simple HP smart control is considered the most effective strategy.

We can observe that despite significant improvements, most of the gains of this final scenario were achieved thanks to the implementation of the microgrid and centralized HP and BESS, and not through the smart control of these assets. This can be observed Figure 26 in, that compares the marginal improvement of implementing smart controls in a collection of decentralized homes, or on a centralized EC. The findings suggest that for most metrics the improvement is greater in a distributed scheme, likely due to lower room for improvement in the community system, as suggested by R. Fachrizal and J. Munkhammar [3].

Table 12: Key metric results for community HP, EV and battery control strategies.

	Decentralized Smart Homes	Baseline E Community	Centralized HP Control	EC with EV Control	Central BESS Control
Energy Deficit [kWh]	486.6	829	828	829	829
Building Min. T [C°]	20.0	21.1	20.5	21.1	21.1
DHW tank Min. T Top Layer [C°]	46.9	59.5	57.4	59.5	59.5
Comfort change Vs. baseline (%)	+18.60 %	-	-6.25 %	0 %	0 %
Self-consumption fraction [%]	26.8	35.1	36.6	38.4	35.2
CO2 emissions [kg]	34 642	33 269	32 443	39 878	33 336
Electricity consumption (Ecs) [kWh]	82 450	82 333	82 298	82 336	82336
Energy supply costs Y1 [SEK]	168 672	151 242	140 780	162 252	143 689
NPV difference compared to reference system [SEK]	-	- 346,588	- 203,778	- 513,235	- 356,406
Peak power import [W]	62 625	41 122	34 385	44 628	33 721
Average cost of electricity [SEK]	2.04	1.84	1.71	1.97	1.75

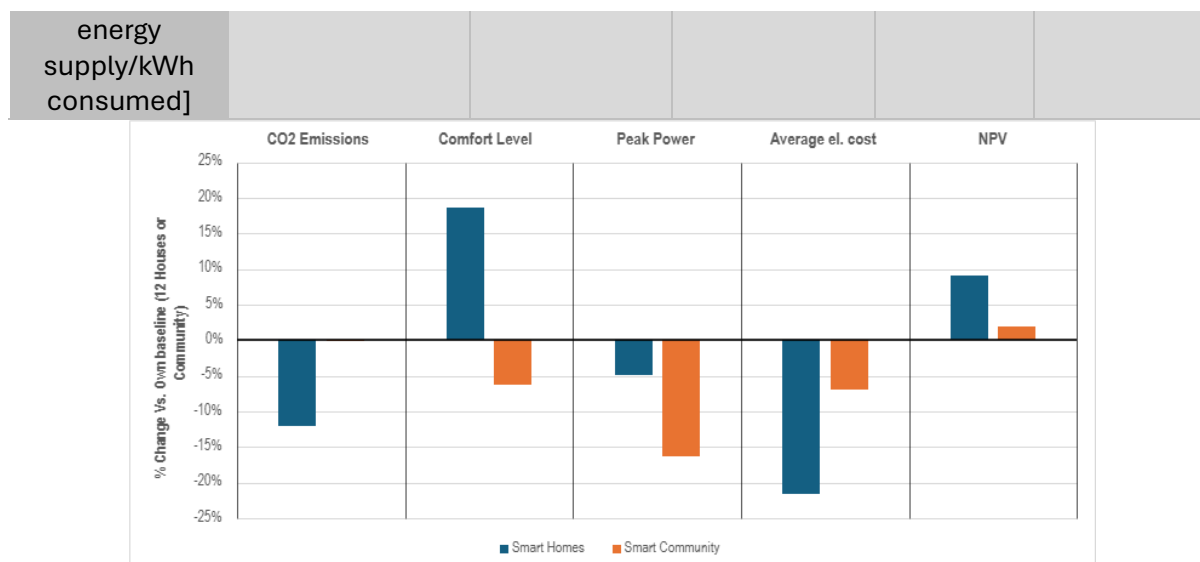


Figure 26: Relative improvement of implementing smart control strategies on top of 12 decentralized homes, or implementing the best performing centralized control (HP) on top of the baseline energy community.

5.5. Community Governance and Stakeholder Impact

As part of a regulatory push from European institutions, ECs were hailed as a natural complement to distributed energy systems, such as the proposed PV arrangements of the neighbourhood studied, and were included in the Clean Energy for all Europeans package[14], released in May of 2019. Through increased SC, and the possibility to reduce grid stress by averaging out the peak power transfers with the grid, institutional bodies identify key benefits of a centralized system that align with the findings of this report. Despite the stated support for ECs, several challenges exist, centred around regulatory uncertainty, complex governance structures, and conflicting interests for stakeholders surrounding the EC.

5.5.1. Energy Community Governance

To allow for the development of a microgrid connecting the various neighbours to a Renewable Energy Community (REC), a single owner of the underlying infrastructure is needed, which will require a step forward from the property developer in taking the role of EC manager. The system must then be open to all homeowners in the new development, which cannot be forced to participate, but rather should be incentivized to participate through attractive benefits over the decentralized alternative. As such, the property developer must consider a competitive business model to recover the 1.6 MSEK increase in CAPEX for a community HP and microgrid, and to compensate their administrative efforts as EC manager.

To enable the full utilization of energy sharing dynamics within the EC, this must attract all neighbours, which will require an offer where every household reduces their costs compared to remaining an isolated household with only a DSO connection. Several methods can be employed, with the most straightforward option being a fee based on the specific consumption of each resident. Without smart controls, the average EC electricity supply costs are 1.84 SEK/kWh, and correspondingly the EC manager should charge a higher multiple of each house's consumption, to cover its costs. However, to offer a competitive plan that benefits all household owners, the property developer should charge an amount that is lower than the costs in the alternative decentralized scenario for each household, which is not the case for households

1003 and 1012 if they opt to implement smart control strategies, yielding an average supply cost of 1.73 SEK/kWh and 1.79 SEK/kWh respectively, shown in Appendix IV. Even adopting the highest performing smart control strategies for the centralized system, supply costs are only lowered to 1.71 SEK/kWh, leaving little margin to charge a premium without discouraging household 1003 from joining the EC, which would in turn increase average supply costs and further reduce the margin for the property developer.

5.5.2. Stakeholder Impact

Whilst legislation forbids DSO's from owning ECs, the DSO is an integral stakeholder in the development of an energy distribution system for the neighbourhood, responsible for the grid connection of all properties. As such, it is essential to cooperate in the design of the EC and seek to design the system that can benefit all actors beyond the individual homeowners, providing advantages to the DSO. Disregarding their interests undermines their support for EC, a critical element to successfully navigate regulatory barriers, as highlighted by A. Riikka's and J. Palm's [13] in their analysis of a failed REC project in southern Sweden.

The first concern of the DSO is a revenue loss from reduced grid purchases, caused by an increase in SC, and from lower peak power surcharge. The yearly energy supply costs for the decentralized neighbourhood ranges between 220 kSEK and 169 kSEK depending on the level of adoption of smart controllers, whereas the baseline EC presents significantly lower costs, between 151 kSEK and 141 kSEK. This reduction is mostly driven by lower peak power surcharges, resulting from a shift from billing all individual peaks and adding them up, to a single bill for the community averaged consumption, which reduces peak power surcharge revenue between 34-62%, whilst the technical reduction at the local substation is only of 19-36 %, depending on the level of smart controller adoption. Several pricing structures could be applied by the DSO to increase costs based on the connection fuse size, charging additional marginal costs for a larger single EC connection than for the individual household lines[15], [16], but this would likely not offset the overall revenue reduction in connection charge that derives from the establishment of an EC.

On the other hand, whilst the peak power reduction is less pronounced than the revenue loss, it remains a significant advantage for the DSO as it reduces the strain on the grid, avoids damage on components like transformers, and reduces the risk of affecting the power quality negatively. This could reduce long term costs for grid expansion and protection. Currently though, the smart controllers were designed for the homeowners and the peak shaving that is done is only based on the load perspective. This is apparent in Table 13, where even with the smart control for the battery, we reduce the peak power export by an insignificant amount of only 400 W.

Table 13: Comparison of present BESS strategies for EC without an inclusion of the DSO

	Baseline – EC	With Battery - 20 MWh without subsidy	With Battery - 20 MWh subsidized
Peak export to grid [kW]	72.0	71.6	71.6
50 % subsidy in absolute price [SEK]	-	-	60,000
NPV difference [SEK]	-	- 9,818.31	50,181.61

For the DSO and for the political actors involved (municipality or the government), it would be therefore of great interest to subsidize these BESS and to play a role in implementing the smart controllers for the storages, to manage the peak export in a better way and to protect the grid. The EC already reduces peak PV exports by 2.7 kW compared to decentralized baseline homes, and one large EC battery programmed with this goal in mind is less expensive to subsidize and operate than 12 individual home batteries. These subsidies could become conditional on implementing peak power export control inside the smart controllers, which would increase the welfare of home owners through a subsidized battery, and simultaneously reduce grid strain for the DSO, benefiting all parties. However, with the current state of affairs high peak PV exports remain a significant challenge for the DSO, and represent an additional obstacle to develop an EC.

The final stakeholder engaged in the development of a new residential environment is undoubtedly the municipality, which as the local representative of public authority is interested in maintaining the resident's comfort, while keeping the costs for the homeowners and for the DSO as low as possible. The subsidies stated in literature are currently around 50 % for BESS for consumers in Sweden. In our case, while applying a subsidy of 50 % the NPV after 20 years increases by 50,182 SEK, if combining it with the HP, this could result in an increase of 200,000 SEK. Consequently, applying any available subsidies to this report's calculations can significantly impact the final recommendations.

6. Conclusion, final proposal and future challenges

In this report, smart controllers for individual households, as well as energy communities and smart controllers for energy communities were investigated for a new property development. The methodology evaluated KPIs to choose the best strategy. These KPIs were based mostly on the needs of inhabitants living in these households, specifically the comfort level and the financial impact, with additional considerations on the environmental impact.

The evaluation was compared on 3 levels. First, on the house 1012, three one-purpose strategies (SC, PA and PS) were applied and compared for each of the three technologies – heat pump, EV and Battery storage. The E-level analysis served for this purpose. The three strategies were later combined for each technology separately and evaluated, now for all households 1001 – 1012. This was done in the C-level analysis. Finally, an EC was established, with a centralized HP and Battery, with 12 individual EVs. For this EC smart controllers were tested for HP, EV, and the possibility of including a Battery was discussed.

6.1. Challenges and disclaimers

Some clear challenges arise from the analysis which was done in this project, which include economic, technical and political or social challenges, that were already partly discussed above.

1. Economic challenges

If assumed, that every household already owns PV, an EV and a heat pump, additional improvements need to be made to install a smart controller. Even if this investment may be small, it wasn't accounted for in the analysis. Additionally, to assume, that every household in the specific community already has this grade of electrification is valid in Sweden, where more than 70 % of individual households own a HP [17] of some kind and over 50 % of new car sales are plug-in vehicles [18], but this certainly doesn't apply for other big countries like Germany, the USA, Canada, etc.

Apart from that, if a home already owns a heat pump, building a smart community on top of that would not be in much interest of the homeowner, as this would result in higher capital costs and a much lower improvement return on investment than the ground-up development considered in this study.

2. Technical challenges

Polysun makes some technological assumptions, that cannot be transferred to real world applications and must be therefore revised further.

The EV has a weekly user profile, making it easier to model, but if trying to implement the smart EV control on a real-world household, the profile changes between weeks, but may also change drastically during the summer holidays, or during Christmas. This could result in higher energy deficits, resulting in discomfort of the inhabitants, when the EVs charge is not enough.

The PA strategy for the HP, with a focus on SH, was implemented mostly for houses with floor heating as already mentioned, but wouldn't work as efficiently for households with radiators, as their heat capacity is much lower than the walls inside the building, and the storage of heat in the air's inertia wouldn't be as effective.

A smart control isn't possible for households without a smart meter. It is possible to neglect this technicality for Sweden, with a nearly full smart meter coverage [21], but this looks very different in other countries, for example Germany, where only 3 % of households are connected to the grid through a smart meter [19].

We assume full knowledge of the yearly electrical and thermal consumption of the building. We also assume full knowledge of the yearly prices. Looking for small differences in thresholds that offer big financial benefits, could result completely differently if applied to a real-world example with much different environments. Changing tank sizes or battery capacity might even result in worse TLCC than when sticking to the baseline, due to high investment costs. Thermal consumption might as well increase, if the houses are older and not well insulated.

3. Political and societal challenges

For the implementation of smart meters, smart controls and energy communities, detailed information about the energy consumption and energy use must be accessible for the DSO, the controls or even for the companies which install and operate these controls. This amount of transparency over private information is not an easy topic and collides with public disagreement or even protests against the implementation of smart meters, smart controls and energy communities [20].

6.2. Final proposal

With all the evaluations done, looking at the established KPIs there is no definite best strategy (as in Table 14, where green are the best values in each row), but considering the feasibility of implementation a decentralized system of smart homes is the recommended solution.

The smart controlled EC is the best strategy if looking at long term costs, emissions and self-consumption, with lowest energy costs and the lowest CO₂ emissions. However, the smart controlled EC has higher capital costs than the smart decentralized homes, and presents significant governance challenges to be managed by the property developer. Given the expected costs and profit margins needed to establish the EC, it is likely not possible to offer a competitive energy price that significantly lowers costs for all residents, which can lead to rejection by certain homeowners, and render infeasible the proposed solution.

An alternative to an EC would be individual smart controls, the *C-level decentralized* case in Table 14, which are easier to implement in the present regulatory scheme with a centralized grid. These controls offer the best TLCC, but they have higher energy consumption costs. The political work for these households is also lower, as every household has an individual energy environment and can make separate choices, making implementation easier, and rendering this the most feasible solution to improve the neighbourhood's energy management.

Table 14: KPI comparison between EC and decentralized households as well as with the use of smart controls

	Baseline – decentralized	C-level – decentralized	Baseline – EC	Smart control – EC
Energy Deficit [kWh]	637.4	486.6	829	828
Building min T° [°C]	20.4	20.0	21.1	20.5
DHW min T° [°C]	48.4	46.9	59.5	57.4
TLCC [SEK]	7,975,368	7,245,272	7,591,860	7,449,050
Investment costs [SEK]	4,320,000	4,349,000	5,920,000	5,920,000
Energy costs in Y1 [SEK]	220,239	168,672	151,242	140,780
CO ₂ emissions [kg]	39,388	34,642	33,269	32,443
Self-consumption fraction [%]	25.9	26.8	35.1	36.6

Even though these KPIs are good initial decision variables, they don't offer the entire perspective on the comparison of the controls, given that there might be regulatory and governmental

restrictions or opportunities, making batteries more feasible, or making EC harder to implement. Climate change and other global risks can also be an important decision aspect, as natural disasters and the frequency of grid blackouts may underscore the importance of self-sufficient neighbourhoods, which is highest in the EC system. However, with the current state of affairs, a decentralized smart neighbourhood is the best solution for a property developer to offer houses with low environmental impact, cheap electricity bills, and comfortable living conditions.

7. References

- [1] 'Heat pumps: phases of transition differ strongly among countries in Europe | Heinrich Böll Stiftung | Brussels office - European Union'. Accessed: Dec. 12, 2025. [Online]. Available: <https://eu.boell.org/en/2025/11/19/heat-pumps-phases-transition-differ-strongly-among-countries-europe>
- [2] 'Europe's electric vehicle market leaders: Denmark, Sweden, and Finland', International Council on Clean Transportation. Accessed: Dec. 12, 2025. [Online]. Available: <https://theicct.org/publication/europes-ev-market-leaders-denmark-sweden-and-finland-aug25/>
- [3] R. Fachrizal and J. Munkhammar, 'Improved Photovoltaic Self-Consumption in Residential Buildings with Distributed and Centralized Smart Charging of Electric Vehicles', *Energies*, vol. 13, no. 5, p. 1153, Jan. 2020, doi: 10.3390/en13051153.
- [4] C. Ioakimidis, D. Thomas, P. Rycerski, and K. Genikomsakis, 'Peak shaving and valley filling of power consumption profile in non-residential buildings using an electric vehicle parking lot', *Energy*, vol. 148, Feb. 2018, doi: 10.1016/j.energy.2018.01.128.
- [5] E. Nyholm, J. Goop, M. Odenberger, and F. Johnsson, 'Solar photovoltaic-battery systems in Swedish households – Self-consumption and self-sufficiency', *Appl. Energy*, vol. 183, pp. 148–159, Dec. 2016, doi: 10.1016/j.apenergy.2016.08.172.
- [6] D. Metz and J. Saraiva, 'Use of battery storage systems for price arbitrage operations in the 15- and 60-min German intraday markets', *Electr. Power Syst. Res.*, vol. 160, Apr. 2018, doi: 10.1016/j.epsr.2018.01.020.
- [7] A. Pimm, T. Cockerill, and P. Taylor, 'The potential for peak shaving on low voltage distribution networks using electricity storage', *J. Energy Storage*, vol. 16, pp. 231–242, Apr. 2018, doi: 10.1016/j.est.2018.02.002.
- [8] 'Sweden's solar tax incentives in 2023: ROT and Green Deduction (grön teknik)', Solarstone. Accessed: Dec. 12, 2025. [Online]. Available: <https://solarstone.com/blog/swedens-solar-tax-incentives-in-2023>
- [9] E. Psimopoulos, E. Bee, J. Widén, and C. Bales, 'Techno-economic analysis of control algorithms for an exhaust air heat pump system for detached houses coupled to a photovoltaic system', *Appl. Energy*, vol. 249, pp. 355–367, May 2019, doi: 10.1016/j.apenergy.2019.04.080.
- [10] D. Fischer and H. Madani, 'On Heat Pumps in smart grids: a review', *Sustain Energy Rev.*, no. 70, pp. 342–57, 2017, doi: <https://doi.org/10.1016/j.rser.2016.11.182>.
- [11] F. Pallonetto, M. De Rosa, F. Milano, and D. Finn, 'Demand response algorithms for smart-grid ready residential buildings using machine learning models', *Appl. Energy*, vol. 239, pp. 1265–1282, Feb. 2019, doi: 10.1016/j.apenergy.2019.02.020.
- [12] J. Clauß, S. Stinner, I. Sartori, and L. Georges, 'Predictive rule-based control to activate the energy flexibility of Norwegian residential buildings: Case of an air-source heat pump and

direct electric heating’, *Appl. Energy*, vol. 237, pp. 500–518, Mar. 2019, doi: 10.1016/j.apenergy.2018.12.074.

[13] A.-R. Kojonsaari and J. Palm, ‘Distributed Energy Systems and Energy Communities Under Negotiation’, *Technol. Econ. Smart Grids Sustain. Energy*, vol. 6, Sept. 2021, doi: 10.1007/s40866-021-00116-9.

[14] ‘Clean energy for all Europeans package completed: good for consumers, good for growth and jobs, and good for the planet - European Commission’. Accessed: Dec. 12, 2025. [Online]. Available: https://commission.europa.eu/news-and-media/news/clean-energy-all-europeans-package-completed-good-consumers-good-growth-and-jobs-and-good-planet-2019-05-22_en

[15] ‘Aktuellt pris för elnätsavgiften i Göteborg’. Accessed: Dec. 12, 2025. [Online]. Available: <https://www.goteborgenergi.se/privat/elnat/elnatsavgiften>

[16] ‘Våra abonnemang och priser’, www.ellevio.se. Accessed: Dec. 12, 2025. [Online]. Available: <https://www.ellevio.se/abonnemang/elnatspriser-privat/>

[17] RISE. (2023). *Heat pumps can bring flexibility to the energy system* | RISE. Accessed: Dec. 12, 2025 [Online] Available: <https://www.ri.se/en/heat-pumps-can-bring-flexibility-to-the-energy-system>

[18] Europa.eu. (2025). *Sweden’s EV Market in January 2025: 14.4% YoY increase* | *European Alternative Fuels Observatory*. Accessed: Dec. 12, 2025 [Online] Available: <https://alternative-fuels-observatory.ec.europa.eu/general-information/news/swedens-ev-market-january-2025-144-yoy-increase>.

[19] Tagesschau.de. (2025). *Page Restricted*. Accessed: Dec. 12, 2025 [Online] Available: <https://www.tagesschau.de/wirtschaft/energiewende-sonnenstrom-smart-meter-100.html>.

[20] Gupta, R. *et al.* (2025) ‘Overcoming barriers and seizing opportunities for smart meters in developing countries: Insights from a large-scale field study in India’, *Energy Research & Social Science*, 122, p. 103996. Available at: <https://doi.org/10.1016/j.erss.2025.103996>.

[21] Huang, Y., Grahn, E., Wallnerström, C.J., Jaakonantti, L. and Johansson, T., 2018. Smart meters in Sweden-lessons learned and new regulations. *Current and Future Challenges to Energy Security*, 177.

Appendix I. Financial Inputs and Modelling Horizon

To properly assess the performance of different controls, we must standardize the costs of all system components, in accordance with Table 15 below. The effective interest rate is considered 5%, being the nominal discount rate presented in the project description, and a 20 year project lifetime is considered. A fixed annual fee of 2000 kr/y, and 1000 kr/kW peak power surcharge are considered. Operational costs are taken from the project description to be only the costs for HP (2000 kr/y), PV (1500 kr/y) and BESS (2000 kr/y).

Table 15: Description of main economic inputs for the Polysun model.

Cost included	Unit price	Service life [years]
Heat pump	200 000 kr	20
Photovoltaics (PV)	150 000 kr	30
EV charger	10 000 kr	10
Home battery	25 000 kr + 6000 kr/kWh	Automatic calculation

Appendix II. Metrics for tracking objective completion

The following are the quantitative results provided by Polysun, which can be used to track the attainment of the three main goals for this project.

General variables

- Component Results / Electric consumers / Electricity consumption (Ecs) [kWh]
 - o It is the sum of;
 - Total consumption of profiles [... / Electricity consumption of the hourly profiles (Epcs)]
 - dynamically simulated components, which are;
 - [... / Electricity consumption of the thermal components (Ethcs)]
 - [... / Electric vehicle consumption (Emobcs)]
- Component Results / Heat Pumo / Number of switch-on times
 - o with a hard limit at 2,000 switch-on times per year

Comfortable conditions

- Results Overview / Energy Deficit [kWh]
 - o It tracks the amount of energy needed to meet the needs of occupants. This value includes thermal needs, so a greater deficit means that the temperature setting has not been reached on more occasions.
- Component Results / Building / Temperature / Minimum value [C°]
 - o With a hard limit at 20.0 C°
- Component Results / Storage tank Potable water tank / Top layer / Minimum value [C°]
 - o This can track whether the hot water demand has been properly supplied

Sustainability

- Results Overview / CO2 emissions [kg]

Costs

- Results Overview / Self-consumption [kWh]
- Results Overview / Total energy consumption [kWh]
 - o It tracks the **thermal energy consumption**
 - o Can be used to check if the system is consuming more or less energy (e.g. through battery losses, better HP management, etc...)
- Results Overview / Total electricity consumption [kWh]
 - o It tracks the **consumption of the profiles and thermal (if enabled)**
- System results / Electrical results (annual values) / Self-consumption fraction [%]
- System results / Electrical results (annual values) / Degree of self-sufficiency [%]
 - o Derived from; Component Results / Electric consumers / Self-sufficiency (ESS) [kWh]
 - It is E_{dcs} (Direct consumption of PV) + E_{bchPV} (PV charge of EV, in EV component results) + E_{bdisPV} + E_{bdisCG}
 - o It is the ratio of Self-sufficiency (ESS) divided by total consumption (Ecs)

- To measure peak power, do NOT check “From external grid / Maximum value (Power) [W]”, since that tracks power at each time-step, but peak P charged is averaged over an entire hour
 - o Instead, access the Results menu and find Tabular Evaluation / Hourly values, then can look for “Electric Grid: From external grid [W] (Efeg)” and copy column into an excel. Then from the excel itself or through a python file, obtain the maximum value of “Electric Grid: From external grid [W] (Efeg)”.
 - o Load curves can be plotted through Python to gain insights into the drivers behind peak shaving
- Profitability: System comparison / Energy supply costs [SEK]
- Profitability: System comparison / NPV difference compared to reference system [SEK]
 - o The NPV difference is analogous to the difference in total lifecycle costs (TLCC)
- Profitability calculation / Costs savings through PV [SEK]

Appendix III. Sensitivity Analysis of DHW Tank Storage Size

III.I Self-consumption (Tank size sensitivity)

Using the strategy SC(SH_only), and increasing the tank size, the results are going to be the same, given that the DHW tank is not used in this strategy. However, below are listed the financial results of the SC(SH_only) strategy with a 200 liter tank and all costs of components. Whilst the best performing self-consumption strategy is not sensitive to storage tank size, an analysis can be done on the DHW strategy for self-consumption, to determine if the tank size increase can improve the economic results.

Table 16: Tank size sensitivity Self-consumption

	Baseline -1012	SC(SH_only) - 200l tank	SC(DHW_only) _V2 – 200l tank	SC(DHW_only) _V2 – 400l tank	SC(DHW+SH) _V2 – 400l tank
Energy Deficit [kWh]	56.7	56.8	42.1	46.8	45.9
CO2 emissions [kg]	4146	4083	4138	4098	4053
Electricity consumption (Ecs) [kWh]	8900	8856	9 013	9 031	9 023
Self-consumption fraction [%]	38.1	39	39.7	40.8	41.8
Degree of self-sufficiency [%]	28.3	29.4	29.3	30.5	31.5
Building Min. T [C°]	20.6	20.3	20.6	20.7	20.5
DHW tank Min. T Top Layer [C°]	48.7	47.8	45	49.4	50.5
Energy supply costs [SEK]	842 204	834 414	859 758	824 416	818 479
NPV difference compared to reference system [SEK]	-	+ 6 311	-18 831	+13 329	+ 17 805

Tank sizes of 500l and 600l yield results with a lower self-consumption and lower NPV difference than a 400l tank, and a tank of 300l yields a more negative result than the base scenario with a 200l tank. The strategy of combining SH and DHW control to maximize self-consumption will now be implemented with the optimal 400l tank. The strategy SC(DHW+SH)_V2 – 400l tank yields the best results of all self-consumption strategies, with the highest DHW tank minimum T°, the lowest energy supply costs, the highest self-consumption (41.8%), and the greatest positive difference in NPV compared to the base scenario.

III.II Price arbitrage (Tank size sensitivity)

A sensitivity analysis for the tank size in PA did not yield significant results, largely due to the strong influence of SH regulation in this strategy, as opposed to the low incidence of DHW regulation towards the final results of HP PA. Consequently, a storage tank of 200 litres yields the best results, providing the lowest CAPEX for a strategy that mainly focuses on SH control.

III.III Peak shaving (Tank size sensitivity)

Adapting the storage tank size for peak shaving, we get the following results

Table 17: Tank size sensitivity peak shaving 1/2

	Baseli ne- 1012	PS(APP_EV _V1) 3,5 kW threshold – 200l tank	PS(APP_E V_V1) 3,5 kW threshold – 300l tank	PS(APP_EV _V1) 3,5 kW threshold – 400l tank	PS(APP_EV _V1) 3,5 kW threshold – 500l tank	PS(APP_EV _V1) 3,5 kW threshold – 600l tank
Energy Deficit [kWh]	56.7	57.3	62	58.7	60.6	60.6
CO2 emissions [kg]	4146	4106	4117	4153	4121	4128
Electricity consumption (Ecs) [kWh]	8900	8819	8809	8803	8819	8814
Self- consumption fraction [%]	38.1	38	37.7	36.8	37.7	37.5
Degree of self- sufficiency [%]	28.3	28.4	27.9	27.1	28	27.9
Building Min. T [C°]	20.6	20.6	20.5	20.6	20.6	20.6
DHW tank Min. T Top Layer [C°]	48.7	47.3	45.5	47.9	48.4	48.8
Energy supply costs [SEK]	842 204	768 552	774 264	774 172	774 315	773 151
NPV difference compared to reference system [SEK]	-	+73 841	+67 202	+ 68 463	+66 211	+67 191
Peak power import [W]	7007	5365	5407	5361	5426	5377

From these results, increasing the tank storage size doesn't seem to have an effect in reducing the peak power, and thus has worse results than the original 200 litre tank. With the best performing tank size increase, the 400 litre tank, the threshold can be lowered to 2,8 kW, and the larger tank can compensate for the increase in hours when DHW heating is interrupted.

Table 18: Tank size sensitivity peak shaving 2/2

	Baseline- 1012	PS(APP_EV_V1) 3,5 kW threshold – 200l tank	PS(APP_EV_V1) 3,5 kW threshold – 400l tank	PS(APP_EV_V1) 2,8 kW threshold – 400l tank
Energy Deficit [kWh]	56.7	57.3	58.7	58.4
CO2 emissions [kg]	4146	4106	4153	4159
Electricity consumption (Ecs) [kWh]	8900	8819	8803	8687
Self-consumption fraction [%]	38.1	38	36.8	35.3
Degree of self-sufficiency [%]	28.3	28.4	27.1	25.9
Building Min. T [C°]	20.6	20.6	20.6	20.5
DHW tank Min. T Top Layer [C°]	48.7	47.3	47.9	47.8
Energy supply costs [SEK]	842 204	768 552	774 172	765 204
NPV difference compared to reference system [SEK]	-	+73 841	+ 68 463	+78 387
Peak power import [W]	7007	5365	5361	5246

Results suggest that with the larger tank, the 2.8 kW threshold provides the greatest peak power reduction, whilst maintaining occupant comfort at an acceptable level. Having lowered the peak to 5246 W, the minimum possible with heat pump operation, this larger tank provides the optimal size to lower costs as much as possible, within an acceptable range for minimum DHW tank top layer T°.

Appendix IV. Heat Pump Control Combined Strategies

By combining the different control strategies in pairs of two, different options were explored to design the optimal combined HP control strategy. By analysing storage tank sizes, and power thresholds for peak shaving, the optimal controls were identified and are presented in table X below, including 400 litre tanks and 2,8 kW thresholds for PS. Furthermore, a net self-consumption threshold of 100W is implemented in the SC+PS control, to reduce the flickering between on and off states during ramp up and ramp down periods for PV production. Ultimately, any pairwise combination of strategies yields lower improvements than a combination of all three, which provides a 4.8% reduction in CO2 emissions, an 11.59% reduction in average costs of electricity, and an increase in comfort of 15%.

Table 19: Results Heat Pump Control Combined Strategies

	Baseline-1012	PA+PS	PA+SC	SC+PS
Energy Deficit [kWh]	56.7	57.9	46.2	47.1
Building Min. T [C°]	20.6	20.4	20.9	20.4
DHW tank Min. T Top Layer [C°]	48.7	49.6	49.3	46.9
Comfort change Vs. Baseline (%)	-	-1.06%	+21.21%	+12.26%
Self-consumption fraction [%]	38.1	36.3	41.2	41.7
CO2 emissions [kg]	4146	4028	4044	3962
Electricity consumption (Ecs) [kWh]	8900	8528	8959	8848
Energy supply costs Y1 [SEK]	20 760	18 024	19 452	18 415
NPV difference compared to reference system [SEK]	-	+40 141	+16 981	+31 583
Peak power import [W]	7007	5246	6440	5246
Average cost of electricity [SEK energy supply/kWh consumed]	2.33	2.11	2.17	2.08

Appendix V. Optimal Controller Impact on Individual Neighbourhood Households

Table 20: Results for optimal HP+EV control of houses 1001 to 1006.

Household	1001	1002	1003	1004	1005	1006
Energy Deficit [kWh]	40.2	39.3	60.1	35	32.7	36.2
CO2 emissions [kg]	3464	2621	3269	2900	2858	2456
Self-consumption fraction [%]	23%	17%	34%	26%	30%	25%
Building Min. T [C°]	20.6	20.7	20.5	20.7	20.1	20.7
DHW tank Min. T Top Layer [C°]	49.8	50	49	48.3	48.6	47.7
DHW tank Max. T Top Layer [C°]	61	61.1	61.9	61.9	61.1	61
Number of switch-on times [-]	1217	1130	2103	1440	1320	1338
Energy supply costs (discounted over 20 years) [SEK]	383705.34	310163.18	349827.56	336593.16	409030.99	333795.89
Energy supply cost difference in the first year [SEK]	-4490.31	-3063.19	-6189.73	-3690.52	-4307.97	-2832.45
NPV difference compared to reference system (discounted over 20 years) [SEK]	66402.49	45015	90350	55220	61493	41000
Average supply costs in first year [SEK/kWh]	2.280440433	2.212357019	1.794559392	2.046848404	2.00983405	2.190360332
Self-consumption [kWh]	1977	1626	3075	2024	2700	2299
Yield Photovoltaics AC [kWh]	8203.2	8698.6	8698.6	8203.2	8698.6	8698.6
Energy supply costs in first year [SEK]	15792.05	12765.3	14397.75	13853.07	16834.37	13737.94
Total electricity consumption (Ecs)	6925	5770	8023	6768	8376	6272

Table 21: Results for optimal HP+EV control of houses 1007 to 1012, and community average.

Household	1007	1008	1009	1010	1011	1012	Total
Energy Deficit [kWh]	31.8	45.4	40.2	38.7	39.6	47.4	486.6
CO2 emissions [kg]	1977	3931	2350	2320	2734	3762	34642
Self-consumption fraction [%]	19%	38%	21%	19%	21%	39.2%	26.8%
Building Min. T [C°]	20.5	20	20.7	20.4	20.7	20.2	20.00
DHW tank Min. T Top Layer [C°]	50.4	48.2	49.5	50	49.5	46.9	46.90
DHW tank Max. T Top Layer [C°]	61.3	61.1	61	61.8	61.4	61.1	61.90
Number of switch-on times [-]	1097	1475	1361	2223	1702	1611	2223.00
Energy supply costs (discounted over 20 years) [SEK]	246686.18	424514.79	289556.84	291486.09	310353.61	412581.51	4098295.14
Energy supply cost difference in the first year [SEK]	-5006.35	-4663.58	-4888.76	-4087.1	-4567.09	-3779.77	-51566.82
NPV difference compared to reference system (discounted over 20 years) [SEK]	73438	66812	73156	60252	67383	55569	756090.49
Average supply costs in first year [SEK/kWh]	2.078785831	1.875243104	2.314920357	2.333516826	2.124254116	1.730408642	2.045753426
Self-consumption [kWh]	1682	3597	1487	1578	1956	3216	27217
Yield Photovoltaics AC [kWh]	8203.2	8698.6	8203.2	8203.2	8698.6	8203.2	101410.8
Energy supply costs in first year [SEK]	10152.79	17471.64	11917.21	11996.61	12773.14	16980.5	168672.37
Total electricity consumption (Ecs)	4884	9317	5148	5141	6013	9813	82450

Appendix VI. Comparison of battery controller results (E&C_Level)

Table 22: Results of different battery controll strategies (relative improvements compared to baseline in %)

	Baseline-1012	SC	PA	PS	Combined
Energy Deficit [kWh %]	56.7	56.8 +0.2	56.9 +0.2	56.8 +0.2	56.8 +0.2
CO2 emissions [kg %]	4146	3951 -4.7	4203 +0.5	4015 -3.2	3,997 -3.2
Self-consumption [kWh %]	3127	3916 +9.6	3295 +2.1	3151 +0.3	3153 +0.3
Battery discharging energy [kWh %]		751	361	77	79
Electricity consumption (Ecs) [kWh %]	8900	8895 -0.1	8895 -0.1	8554 -3.9	8553 -3.9
Peak power import [W %]	7007	6606 -5.7	6606 -5.7	4764 -32.0	4754 -32.2
Self-consumption fraction [% %]	38.1	44.0 +9.6	40.2 +2.1	38.4 +0.3	38.4 +0.3
Energy supply costs [SEK %]	504,420	488,358 -3.2	435,555 -13.7	437,967 -13.2	435,555 -13.7
NPV difference compared to reference system [SEK %]	-	-38,539 -5.2	-39,945 -5.4	-9,013 -1.2	-7,576 -1.0
Average cost of electricity [SEK energy supply/kWh consumed %]	2.33	2.23 -4.6	2.26 -3.1	2.11 -9.7	2.10

Appendix VII. Different control logic for the HP in each of the 12 households

	DHW dT – LPT [K]	DHW dT – HPT [K]	Hot water tank size [l]
Household 1001	5	5	200
Household 1002	5	5	200
Household 1003	6	4	200
Household 1004	6	5	200
Household 1005	6	4	500
Household 1006	5	5	200
Household 1007	5	5	200
Household 1008	6	4	500
Household 1009	5	5	200
Household 1010	6	4	200
Household 1011	6	4	200
Household 1012	6	4	400